

Nondestructive Evaluation Sciences Branch

Eddy Current Inspection for Detection of Fully Embedded Flaws

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Nondestructive Evaluation Sciences Branch

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Overview

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- Eddy Current testing is commonly used to inspect for surface breaking flaws in metallic structures.
- Low frequency eddy current techniques have been developed to inspect for far surface flaws when access to both surfaces of the structure is limited
 - Airframe fuselage and wings
 - Pressure vessels
 - Tubing and pipelines
- Overlapping a high frequency surface detection with a low frequency far side inspection would appear to provide a method for detecting embedded flaws in the material under test.
- The eddy current response for an embedded flaw is difficult to determine by standards and is likely different from a back-surface flaw at the same depth due to changing constraints on the induced current.
- Electromagnetic modeling can be utilized to determine response from embedded flaws.
 - Models can be experimentally verified for surface flaws on the near and far surfaces.
 - Model extrapolation to embedded flaws will then predict experimental response.
 - Techniques are in work for limited experimental confirmation on embedded defects.

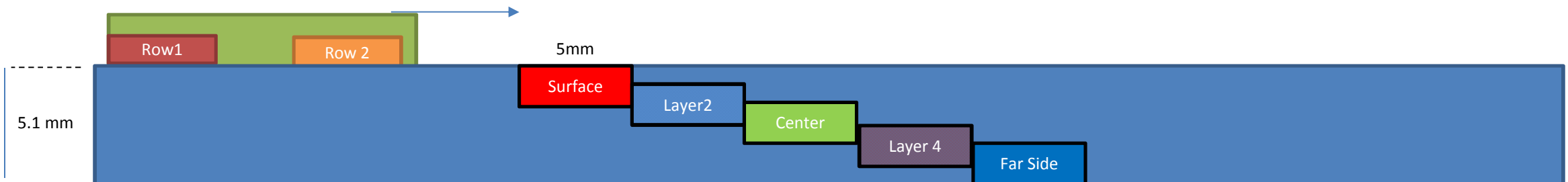


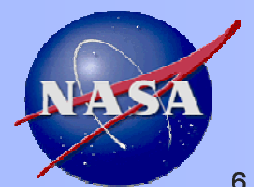
Preliminary Modeling of Eddy Current Inspection for Embedded Flaws.

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- VIC-3D Eddy Current NDE Software used to model Eddyfi ECA-IFG-056-050-048 eddy current coil array response over notches at different depth in Ti-6Al-4V plate.
 - Eddyfi ECA-IFG-056-050-048 modeled at 50 kHz drive frequency. Coil heights assumed at 1.25mm and winding count assumed at 500 turns/coil.
 - Flaw modeled as rectangular notch, 5 mm long x 1.7 mm deep x .25 mm wide.
- Eddy Current probe response modeled as probe scanned in C-Scan fashion at flaw location.
 - 10 mm x 40 mm scan area modeled with 0.5 mm step size.
 - Response calculated both for Axial and Transverse coil configurations.
 - Model can be calibrated against experimental results for known surface and back surface flaws and then used to predict response for embedded cracks.

Eddyfi ECA-IFG-056-050-048

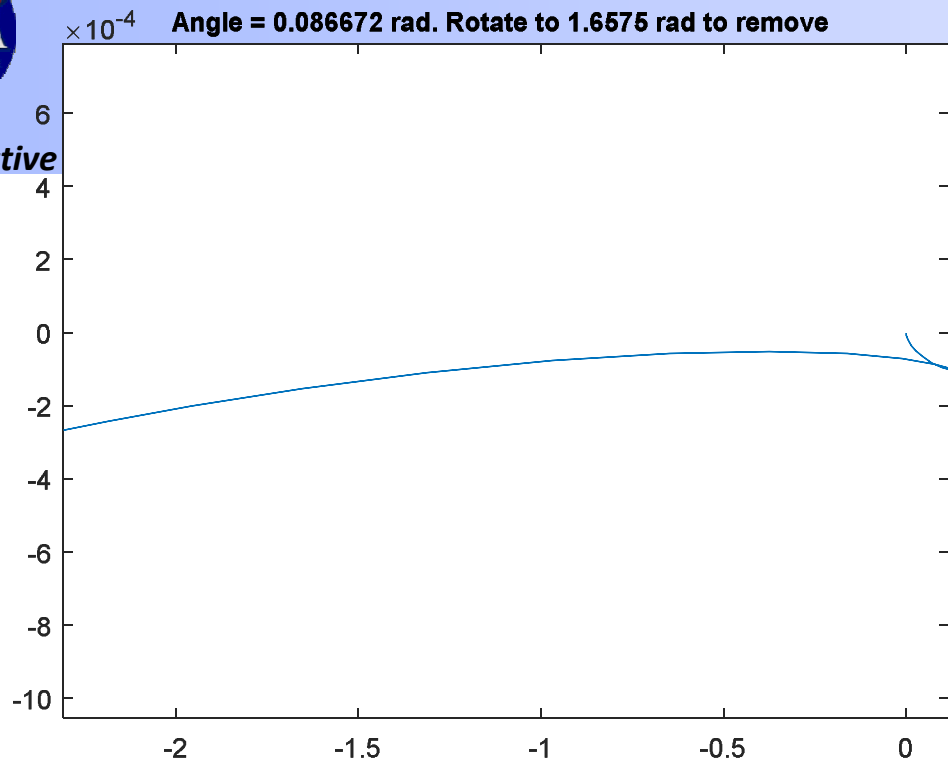




Channel 1

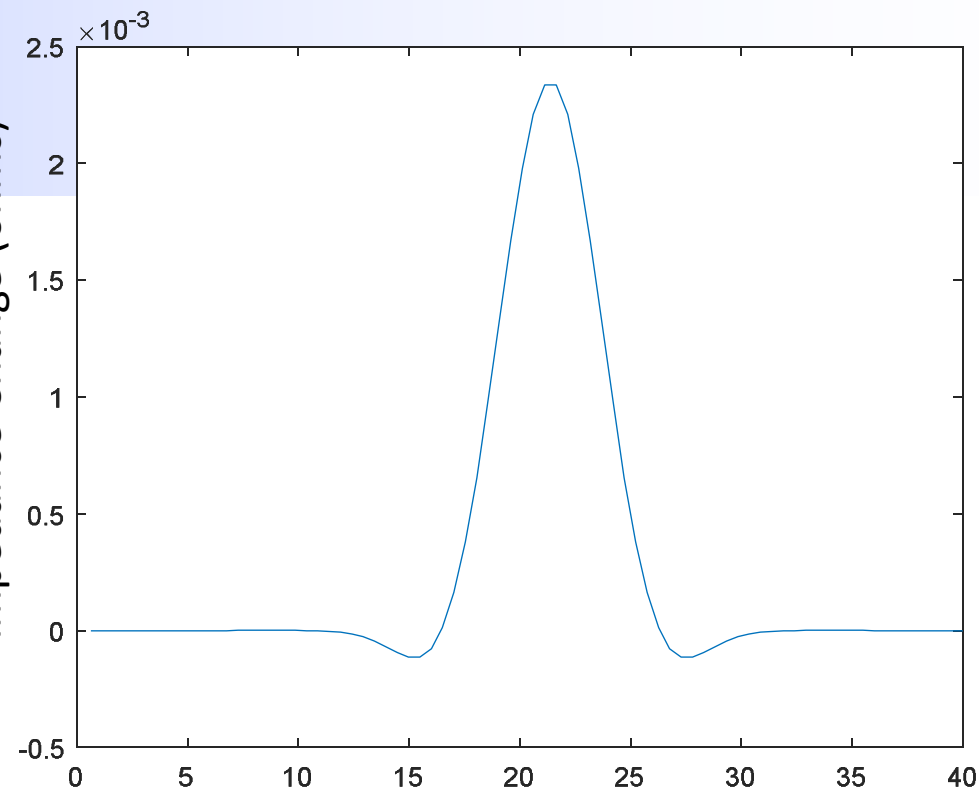
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Reactance (ohms)

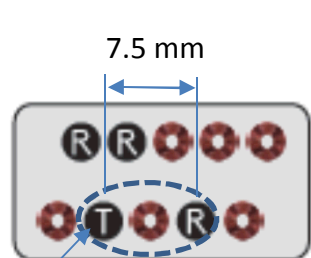


Resistance (ohms)

Impedance Change (ohms)

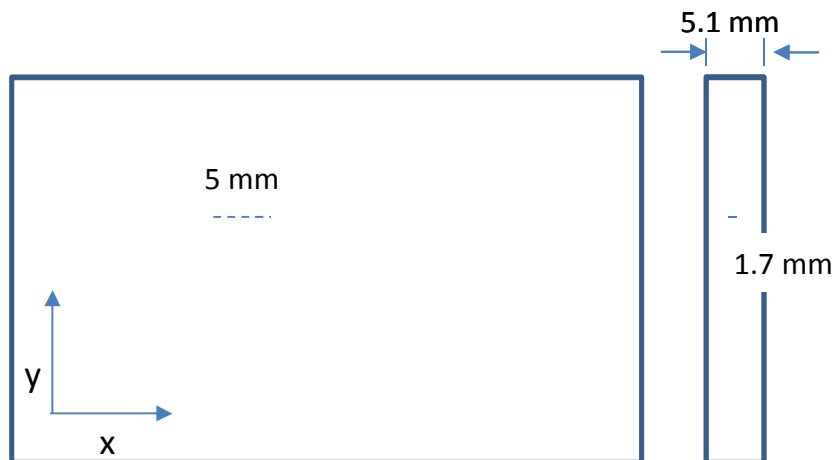


Position (mm)

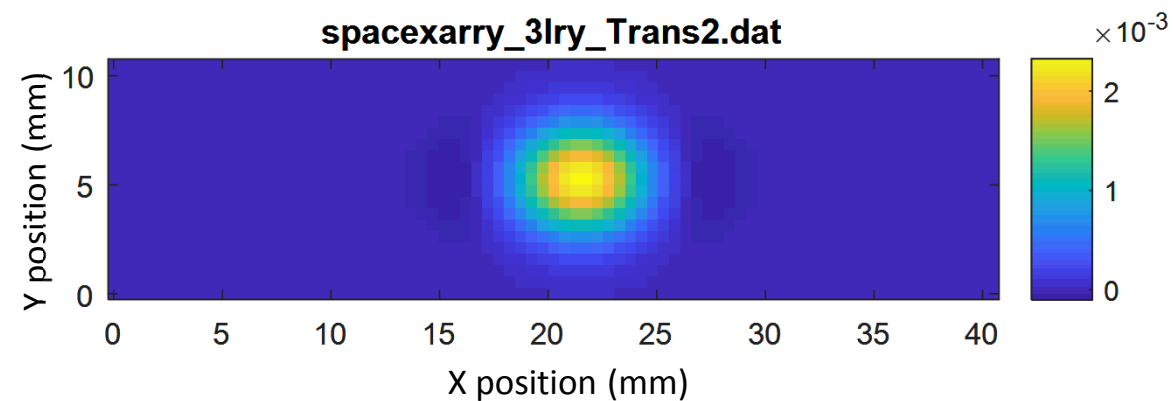


Long, single driver

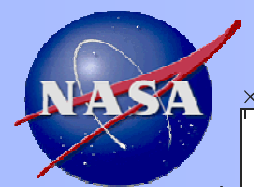
3.5 mm dia



spacexarry_3lry_Trans2.dat



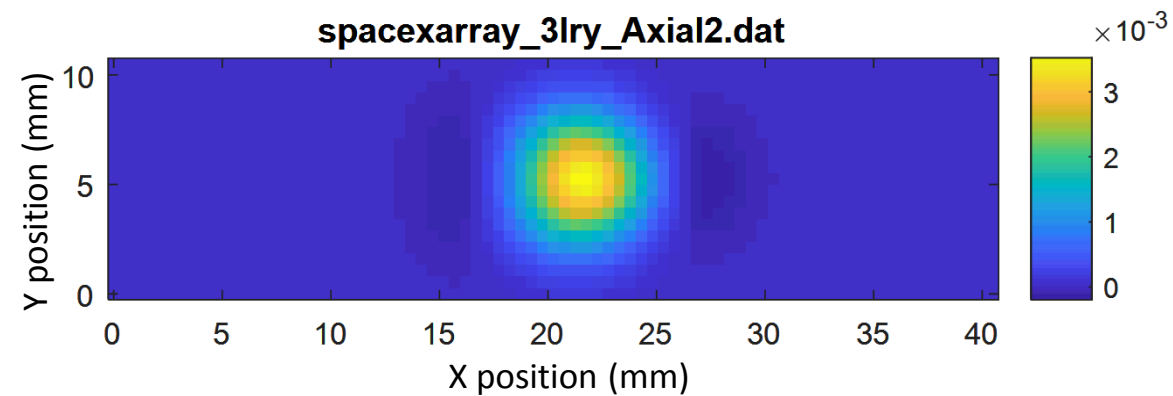
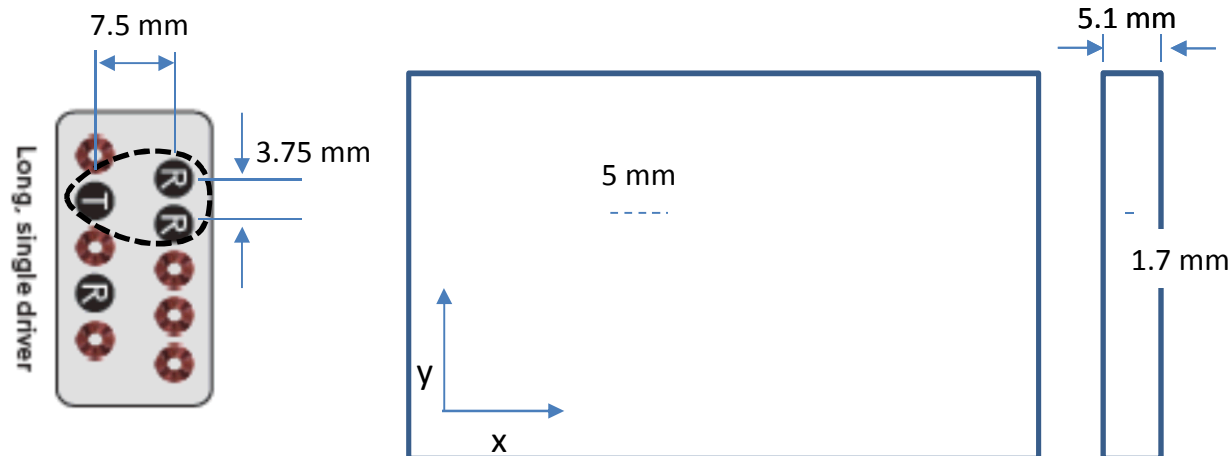
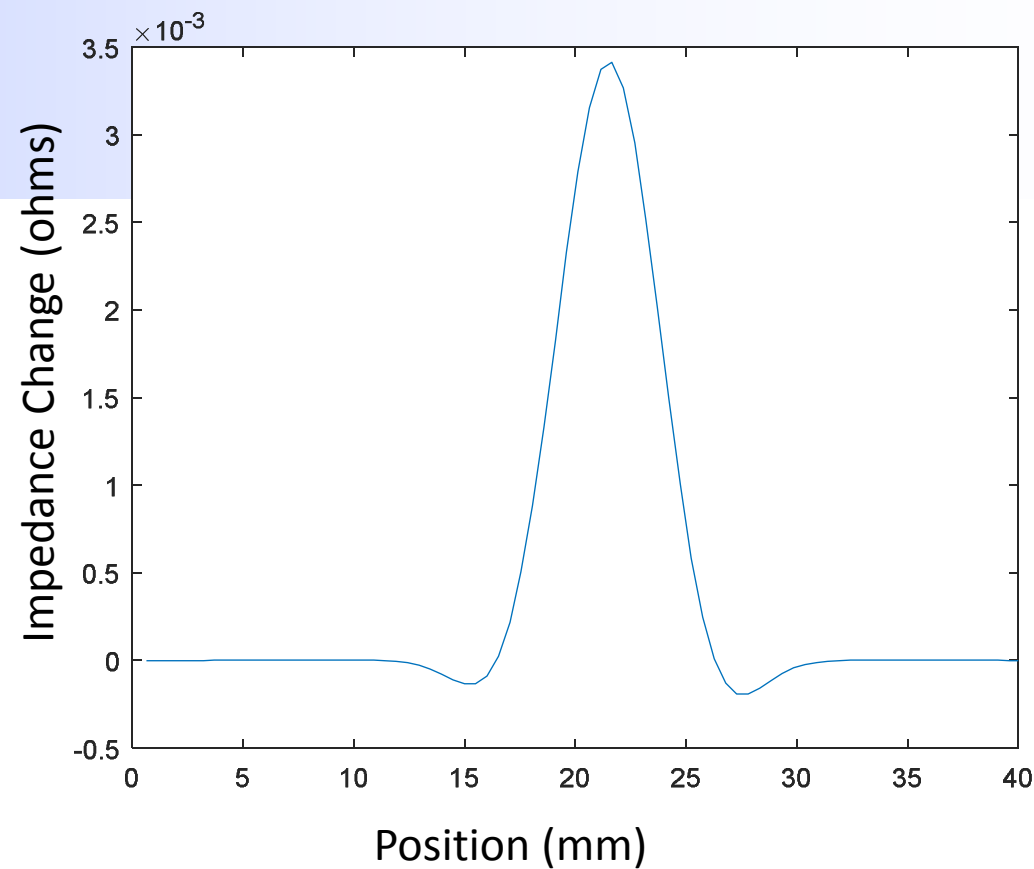
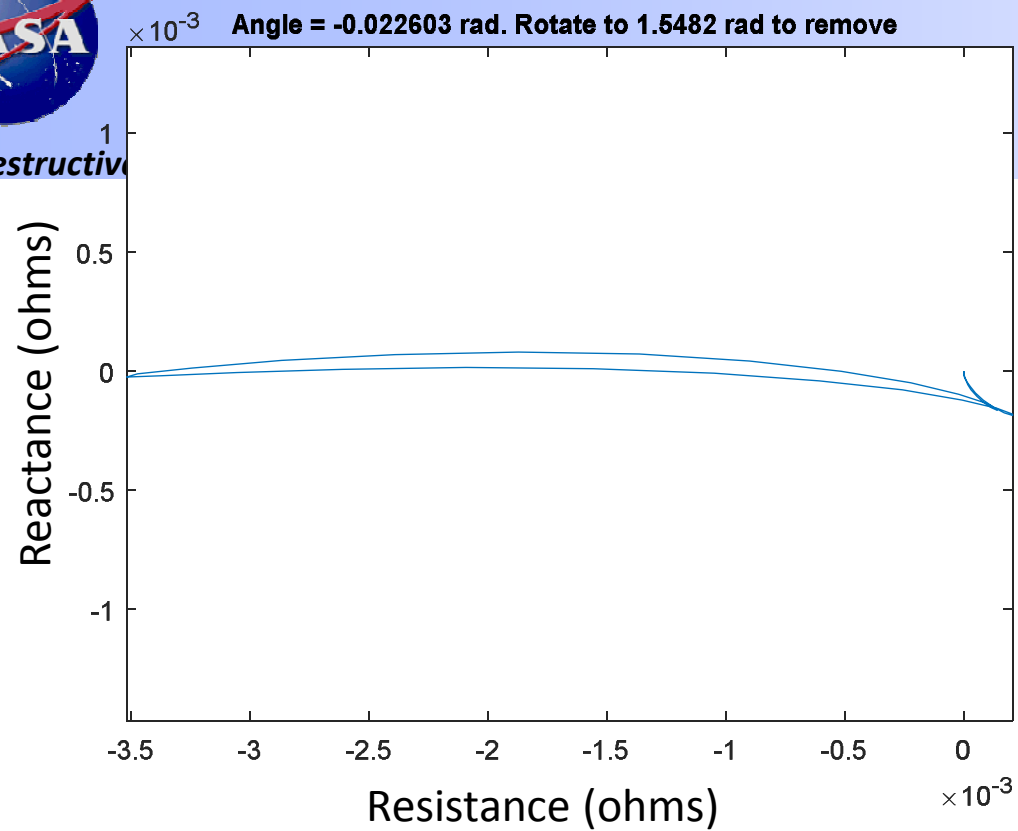
Eddy Current C-scan, flaw response rotated vertical



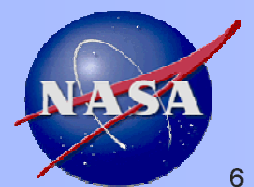
Channel 1

Angle = -0.022603 rad. Rotate to 1.5482 rad to remove

Nondestructive



Eddy Current C-scan, flaw response rotated vertical

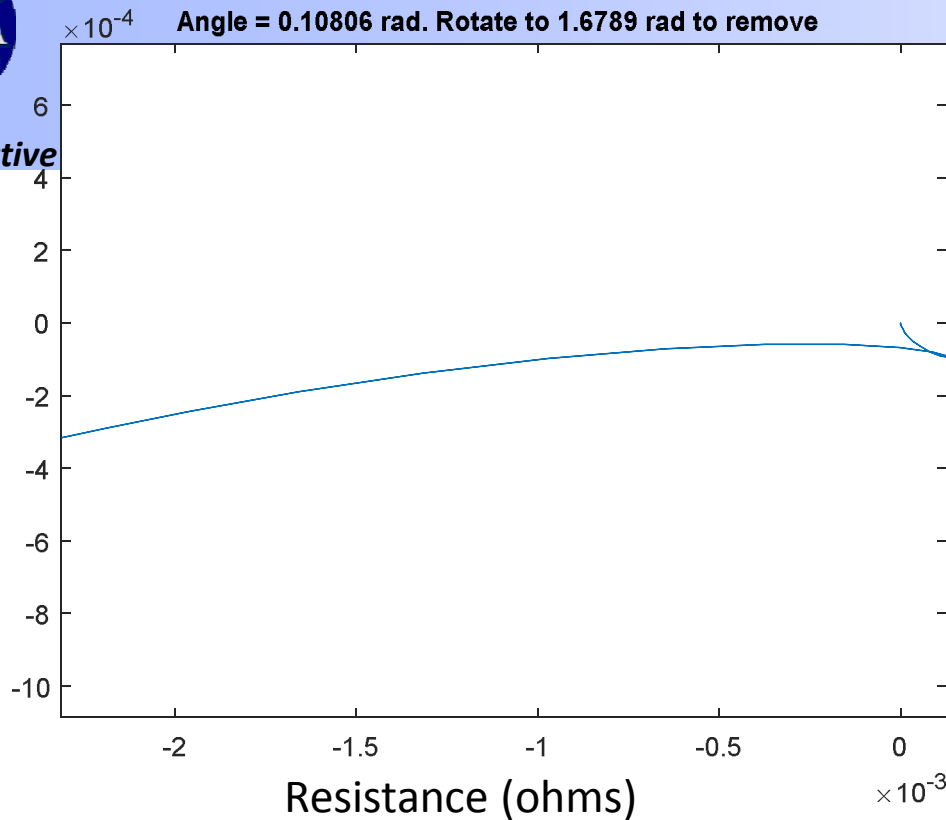


Channel 1

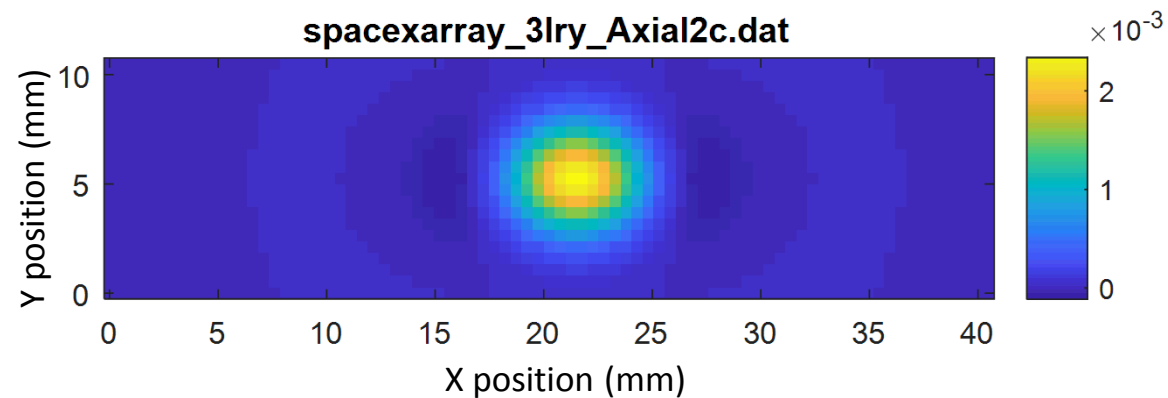
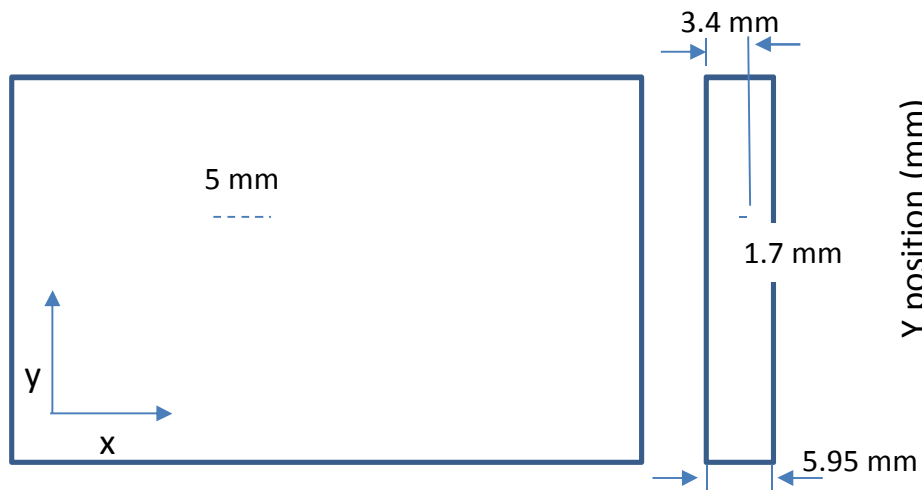
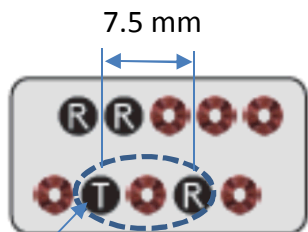
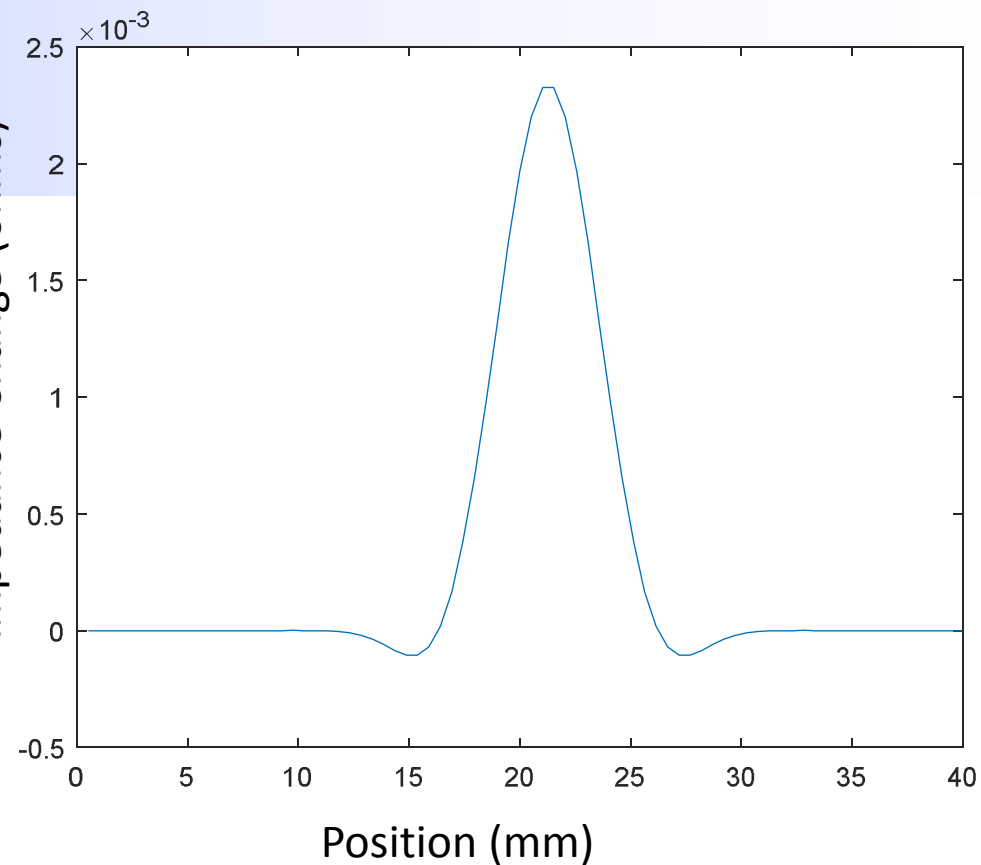
Angle = 0.10806 rad. Rotate to 1.6789 rad to remove

Nondestructive

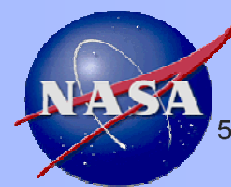
Reactance (ohms)



Impedance Change (ohms)



Eddy Current C-scan, flaw response rotated vertical

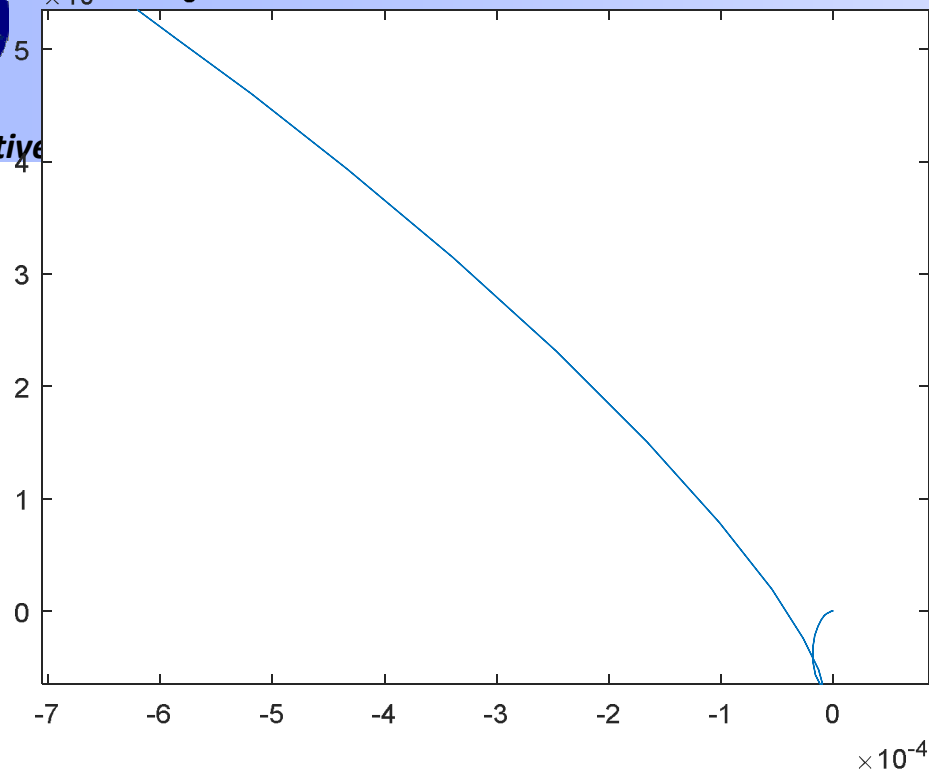


Nondestructive

Channel 1

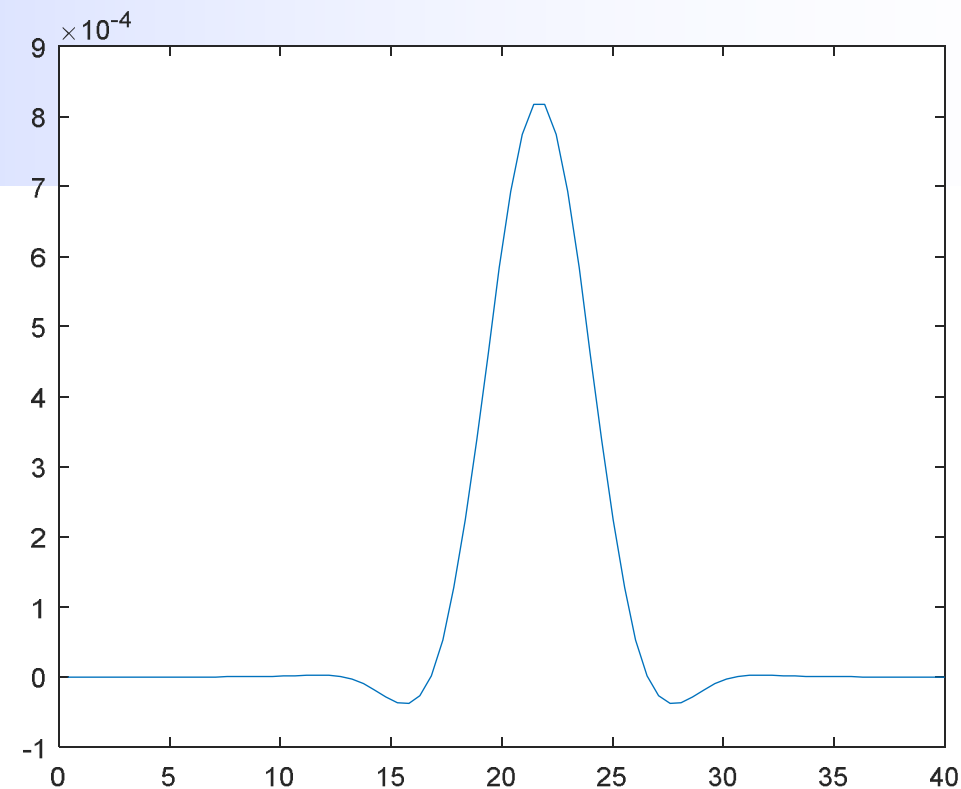
Angle = -0.74328 rad. Rotate to 0.82752 rad to remove

Reactance (ohms)

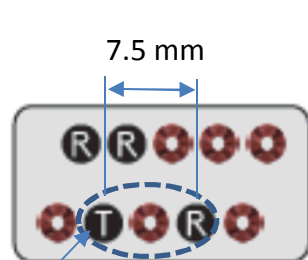


Resistance (ohms)

Impedance Change (ohms)



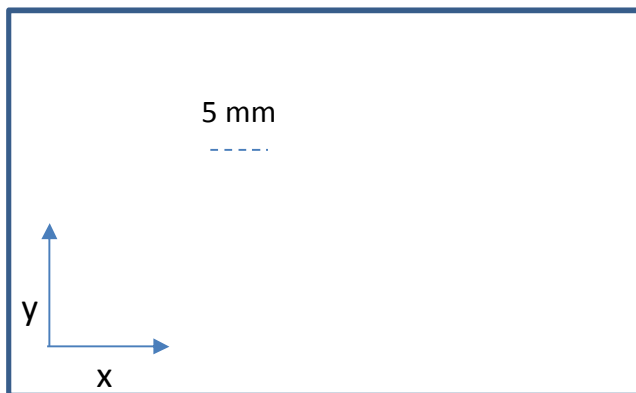
Position (mm)



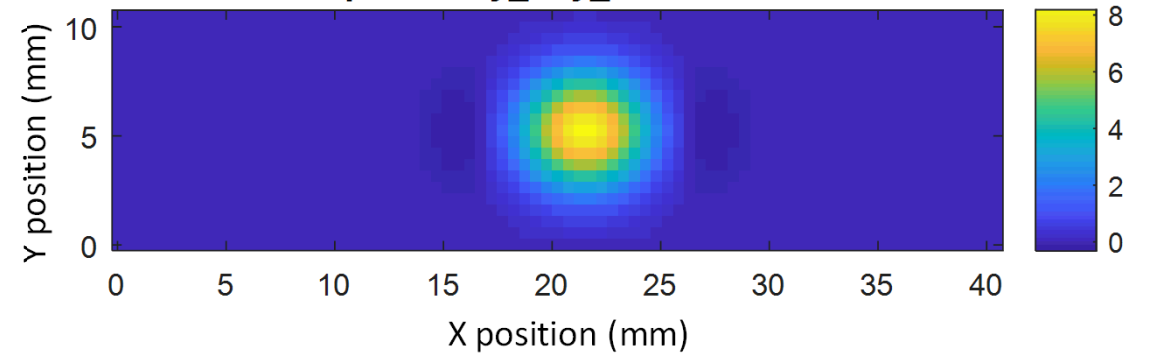
5 mm

5.1 mm

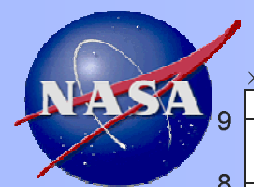
1.7 mm



spacexarry_4lry_Trans2.dat



Eddy Current C-scan, flaw response rotated vertical

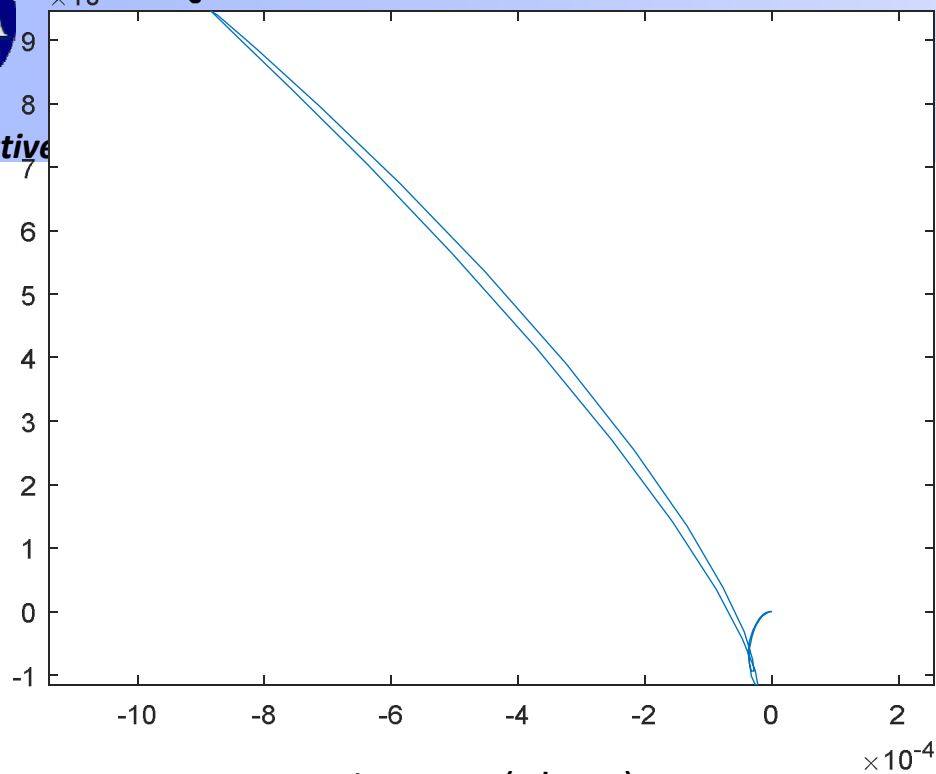


Nondestructive

Channel 1

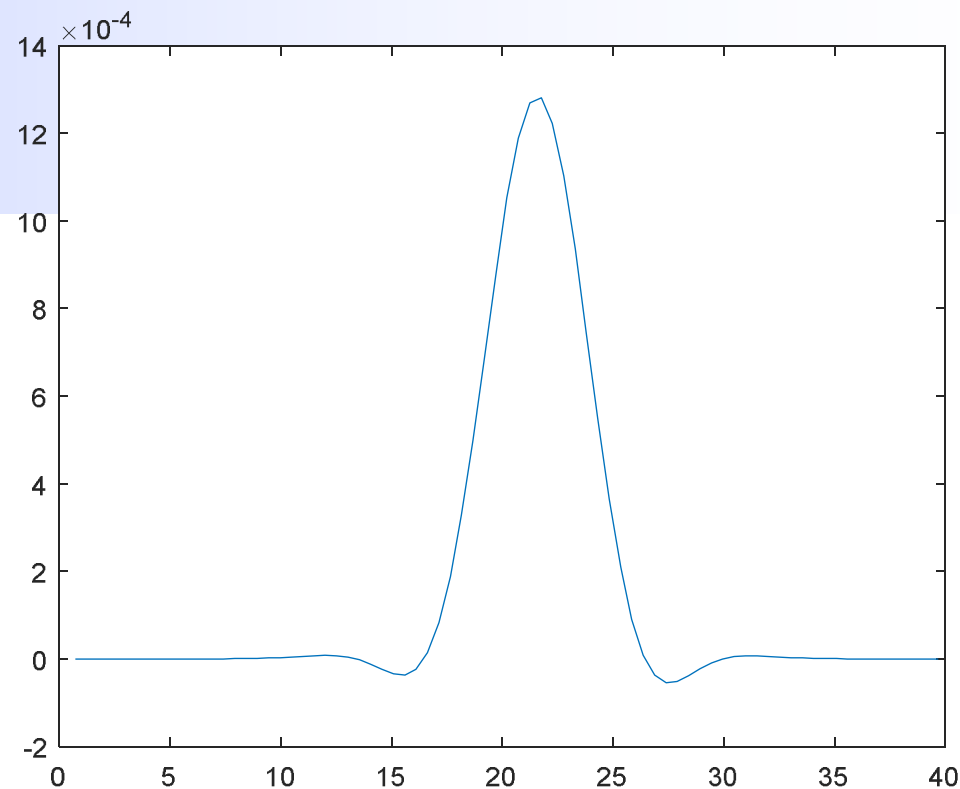
Angle = -0.85032 rad. Rotate to 0.72048 rad to remove

Reactance (ohms)

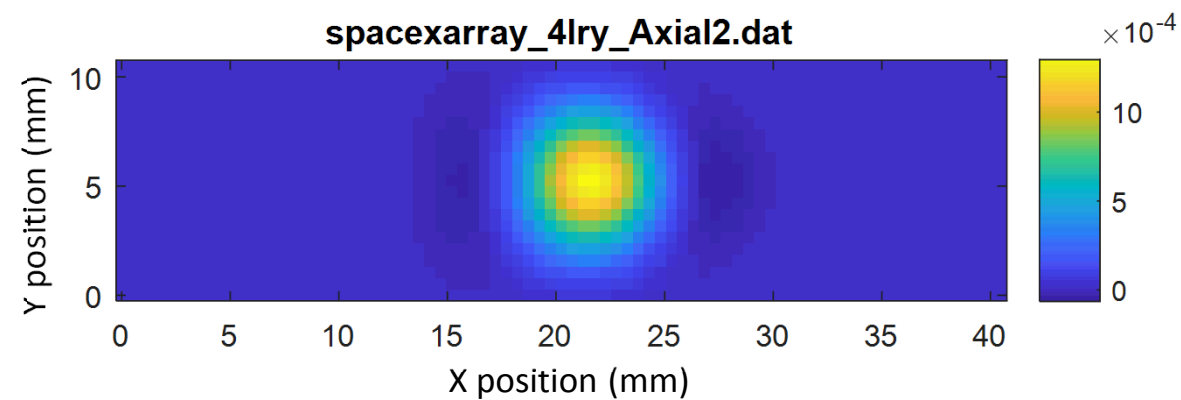
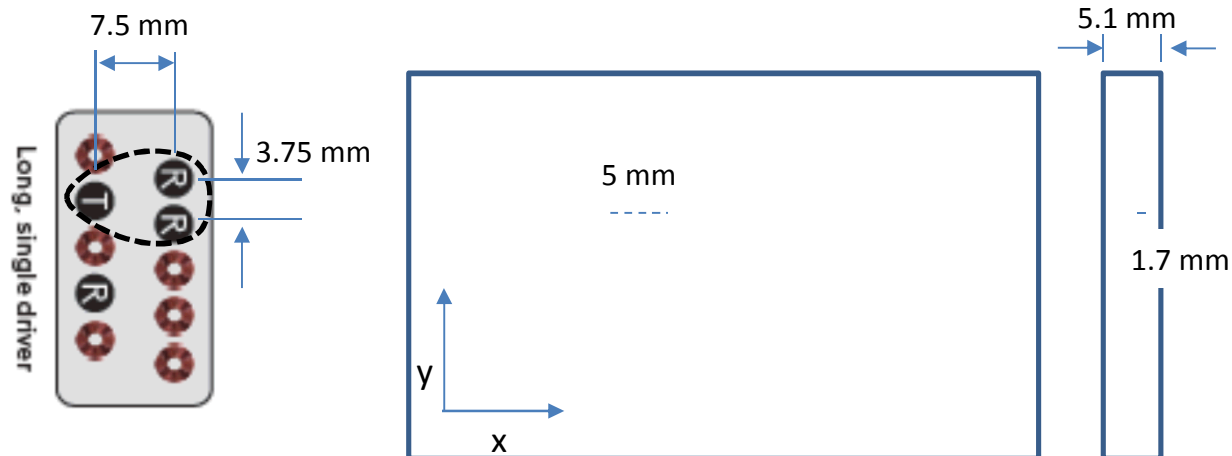


Resistance (ohms)

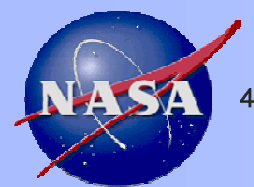
Impedance Change (ohms)



Position (mm)



Eddy Current C-scan, flaw response rotated vertical

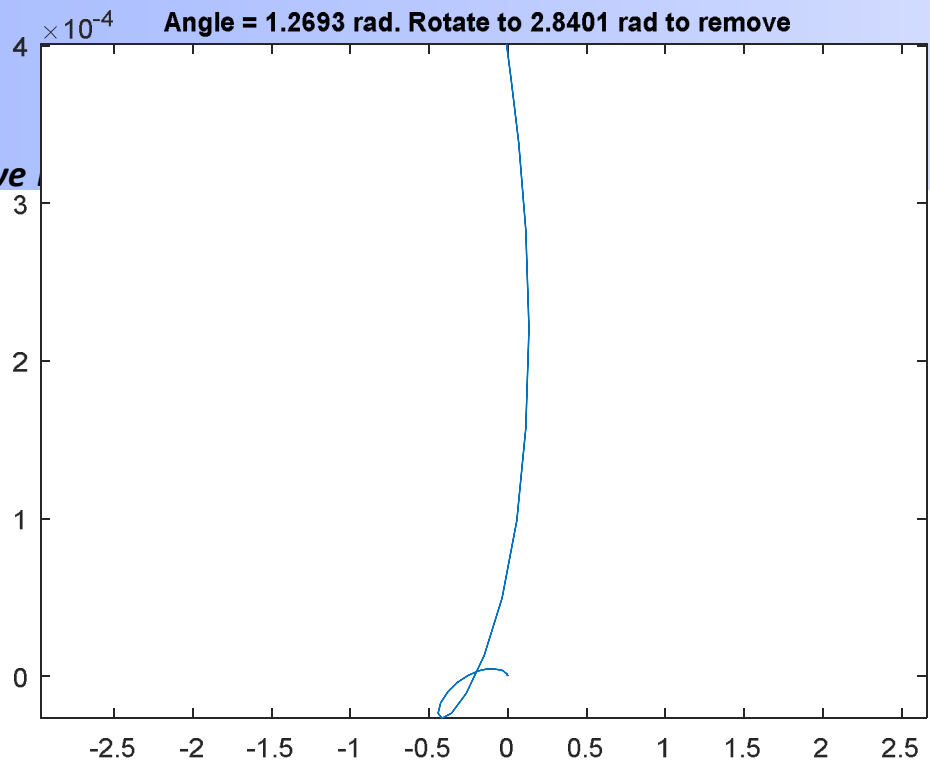


Nondestructive

Channel 1

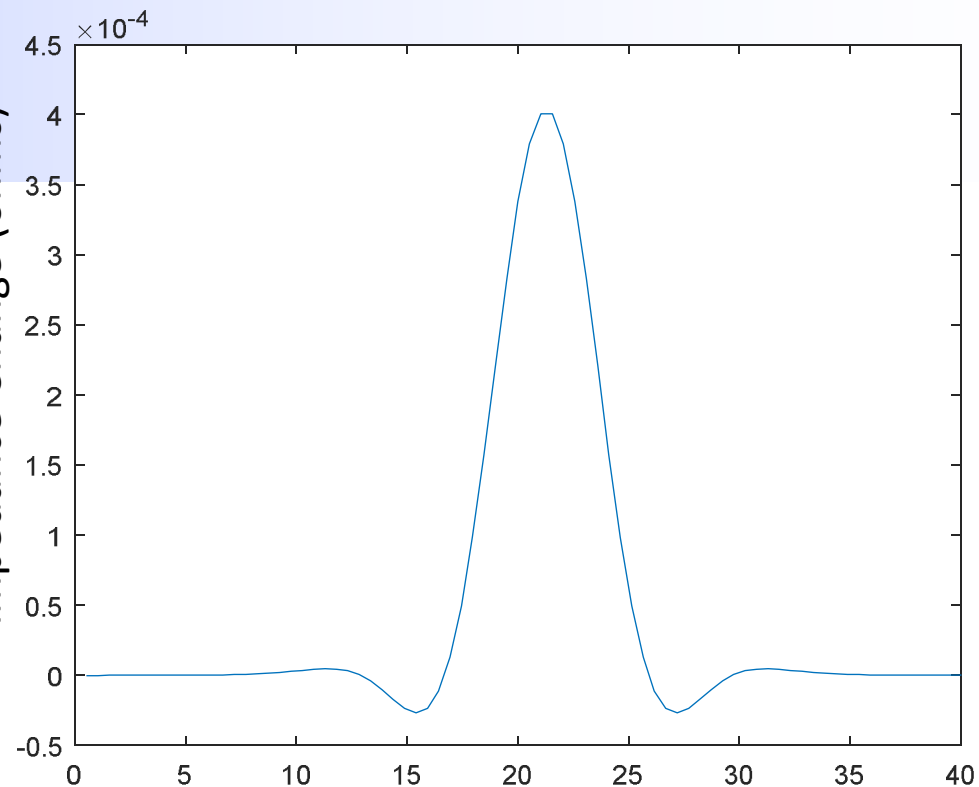
Angle = 1.2693 rad. Rotate to 2.8401 rad to remove

Reactance (ohms)

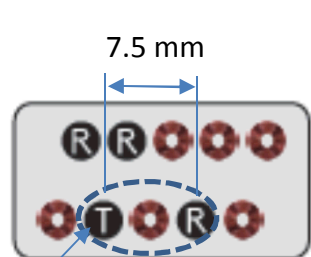


Resistance (ohms)

Impedance Change (ohms)

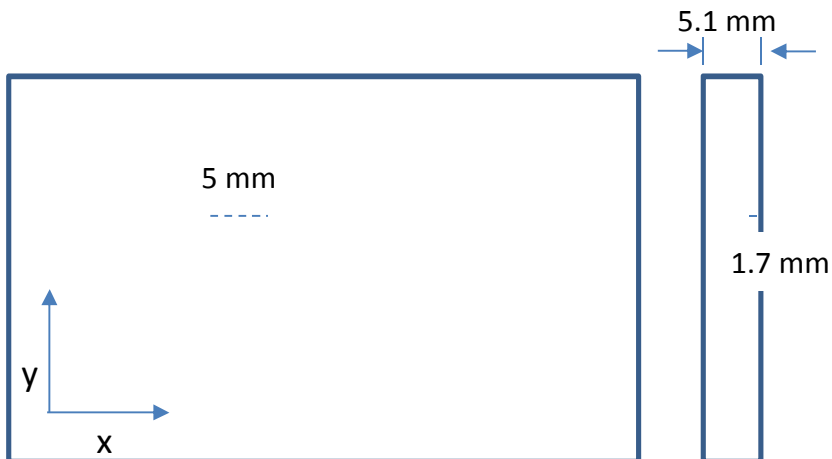


Position (mm)

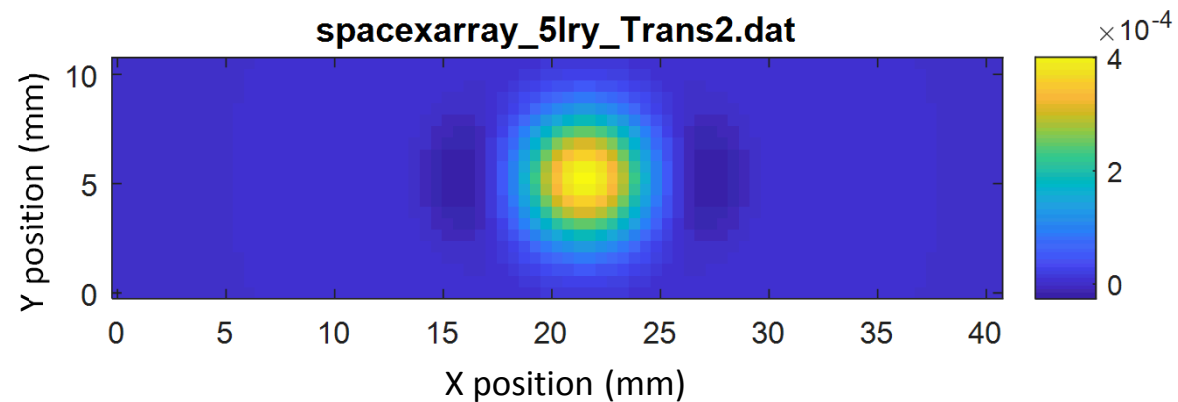


Long, single driver

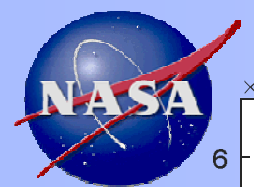
3.5 mm dia



spacexarray_5lry_Trans2.dat



Eddy Current C-scan, flaw response rotated vertical

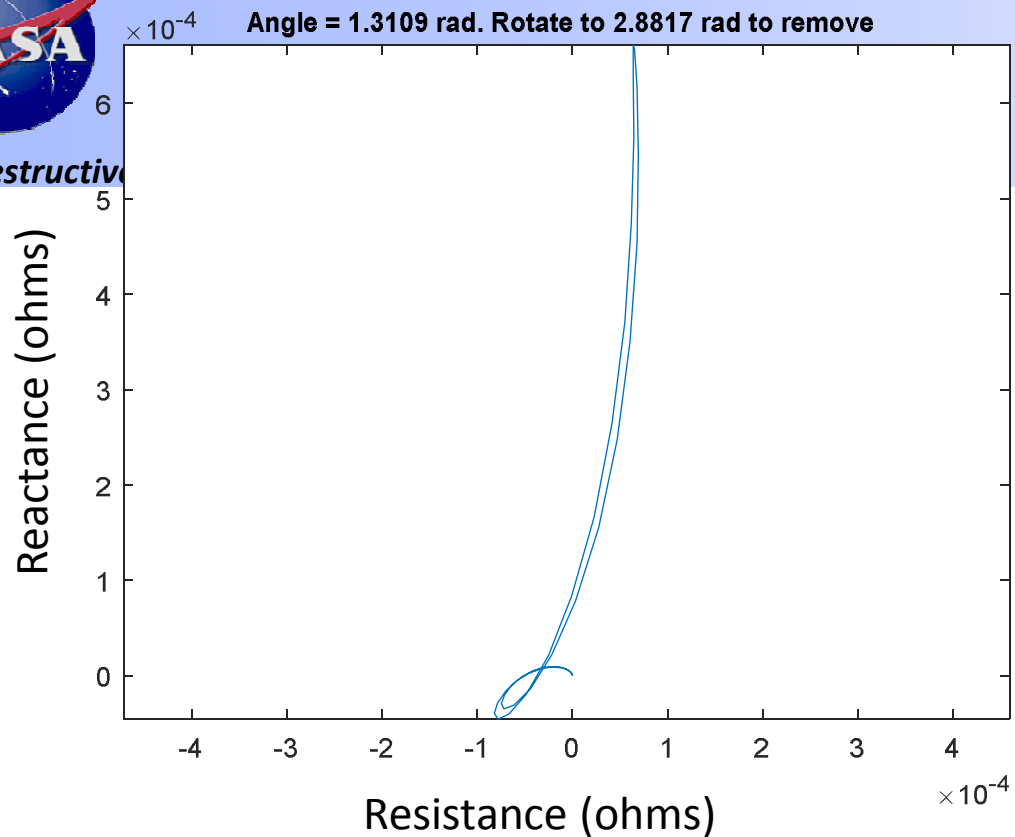


Channel 1

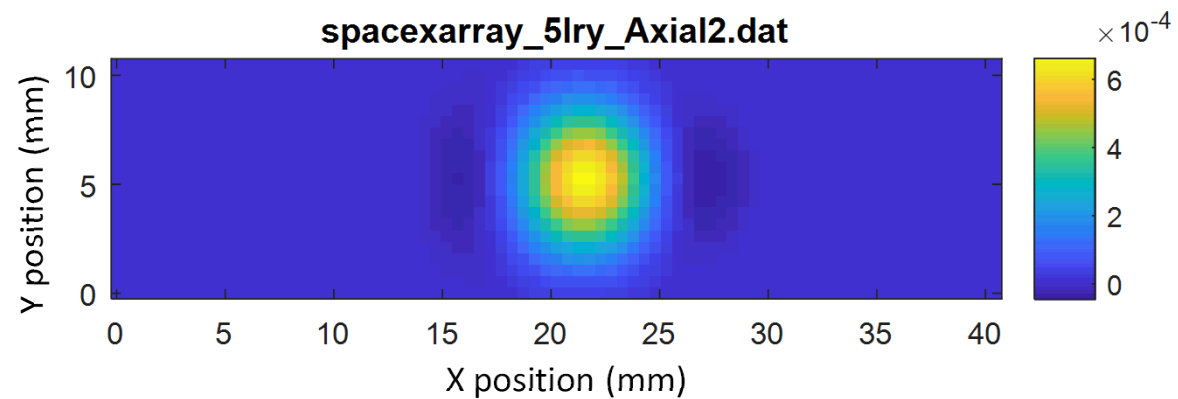
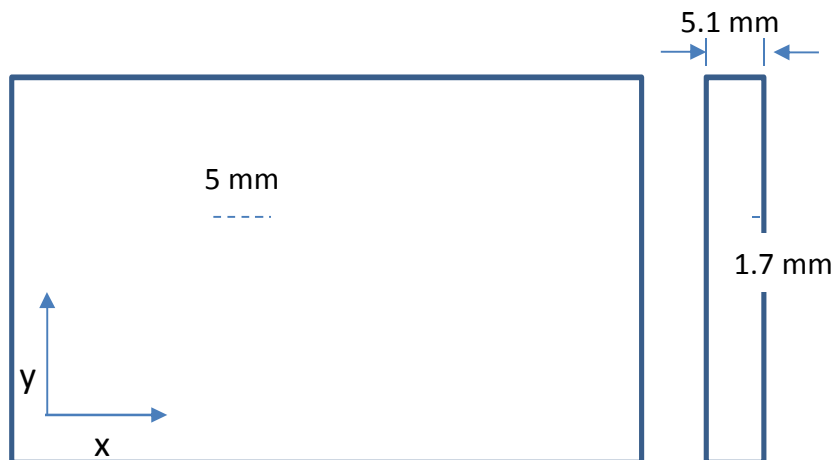
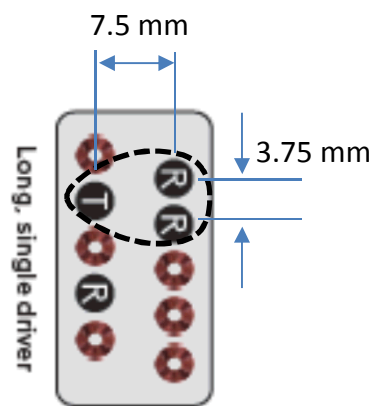
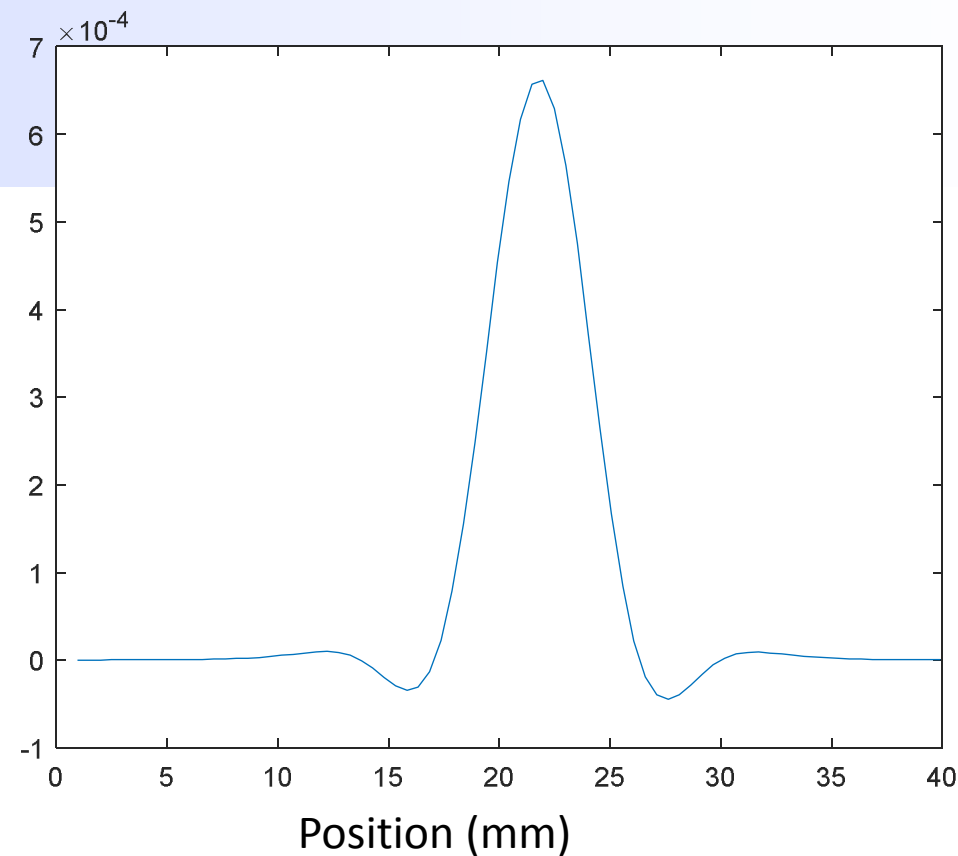
Angle = 1.3109 rad. Rotate to 2.8817 rad to remove

Nondestructive

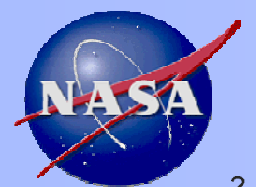
Reactance (ohms)



Impedance Change (ohms)



Eddy Current C-scan, flaw response rotated vertical

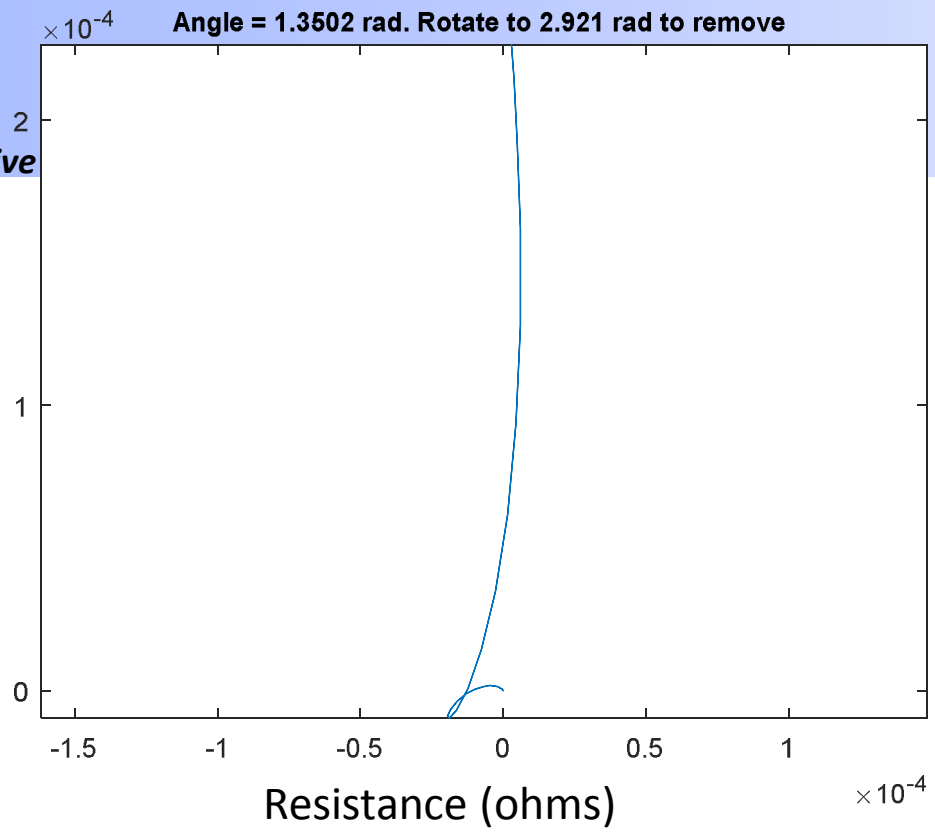


Nondestructive

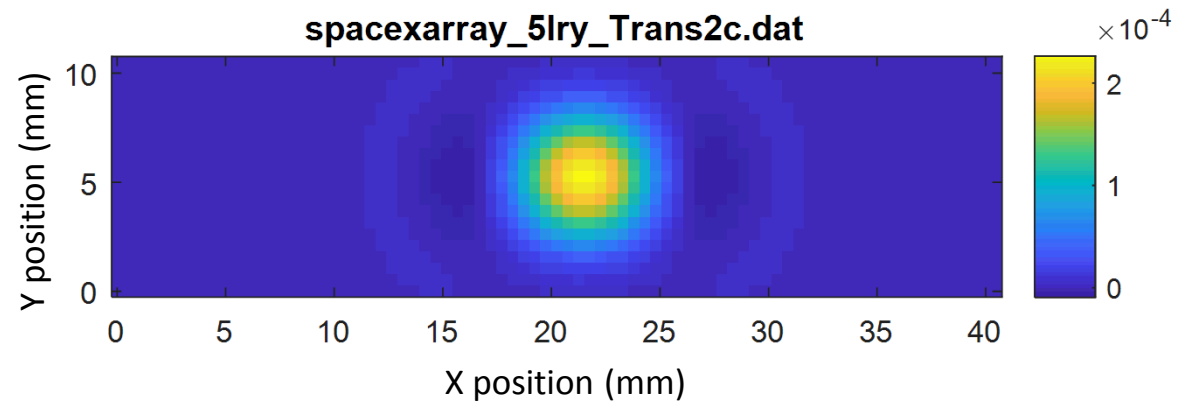
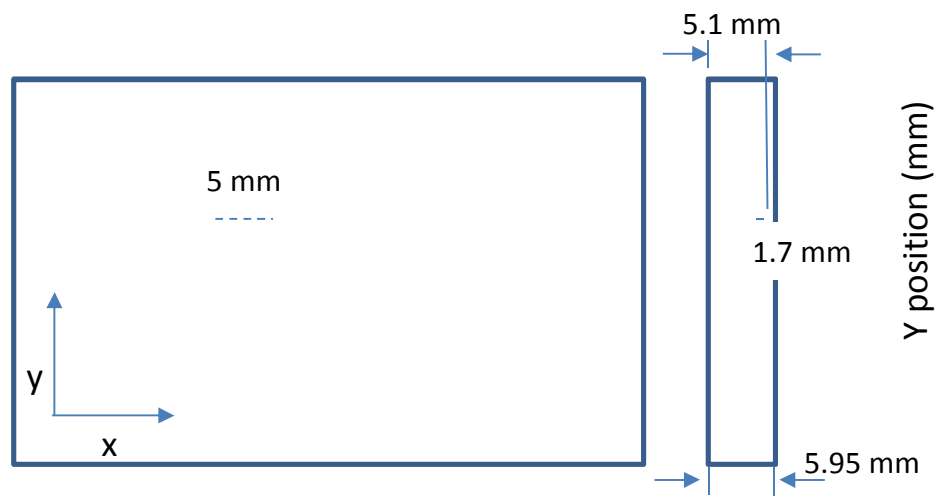
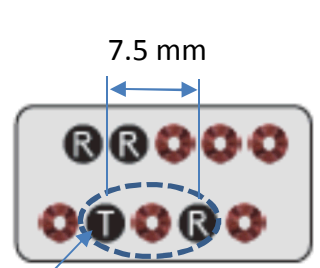
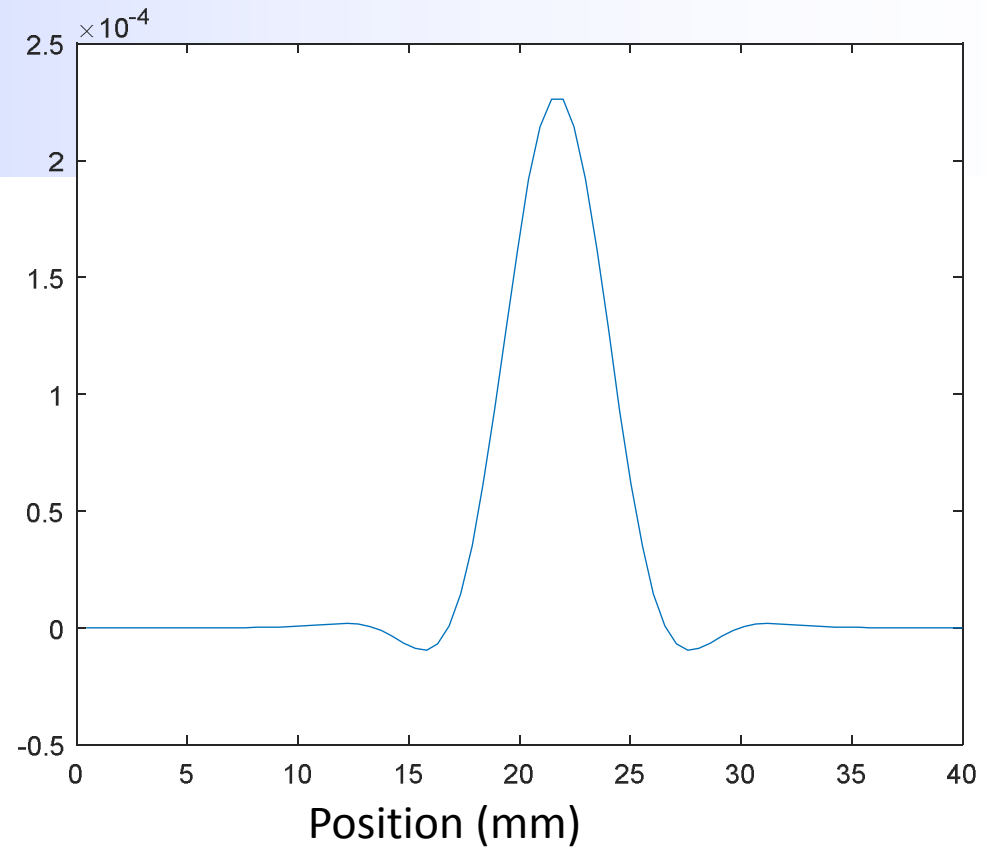
Channel 1

Angle = 1.3502 rad. Rotate to 2.921 rad to remove

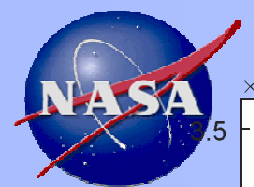
Reactance (ohms)



Impedance Change (ohms)



Eddy Current C-scan, flaw response rotated vertical

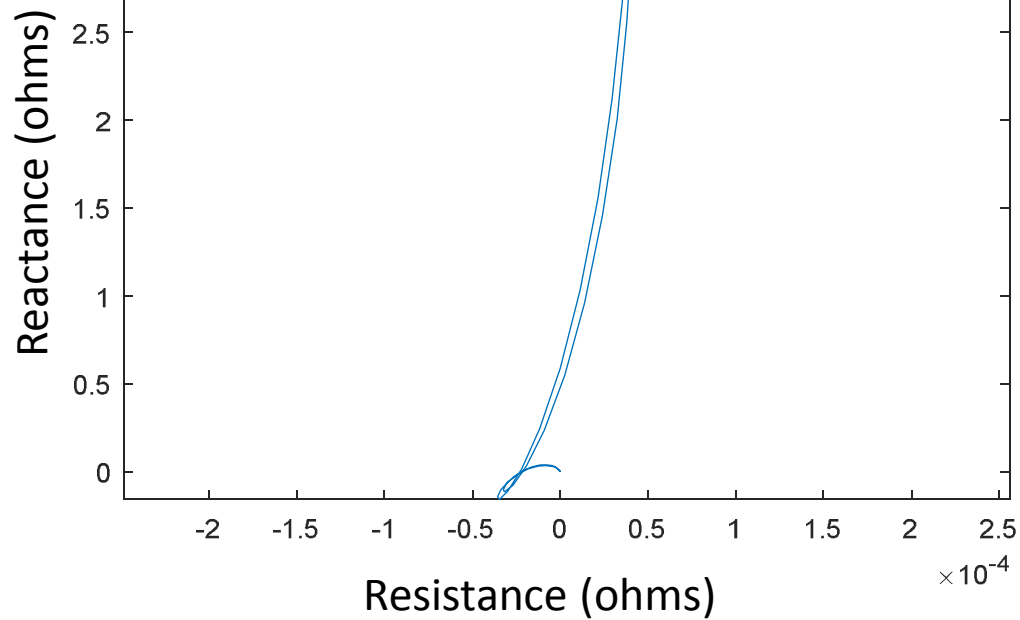


Channel 1

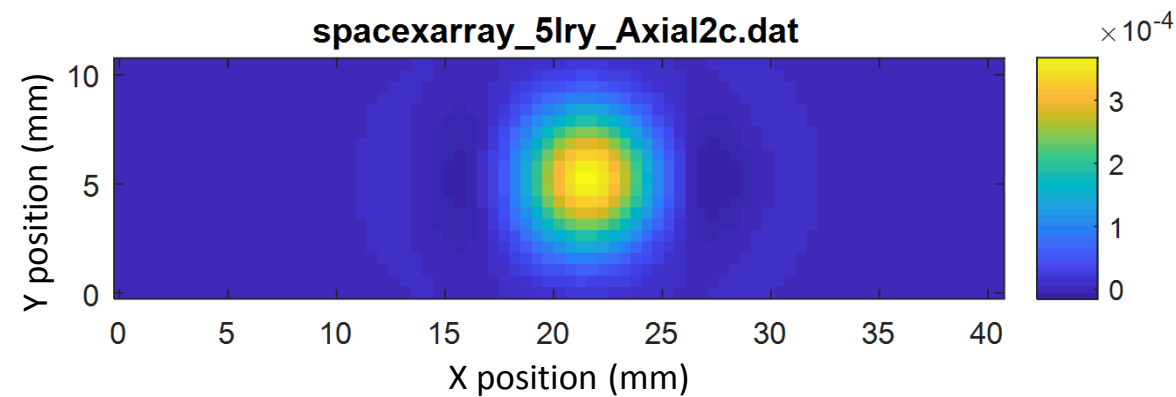
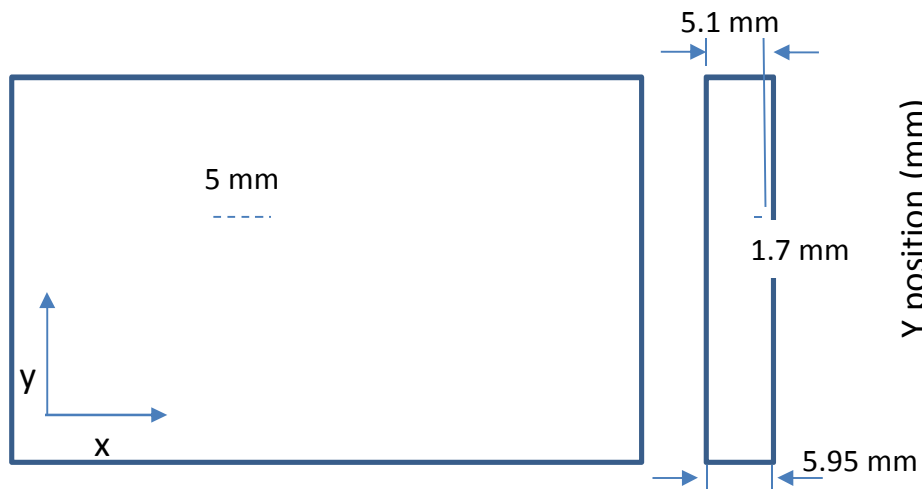
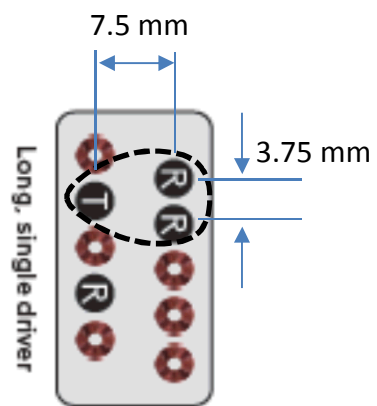
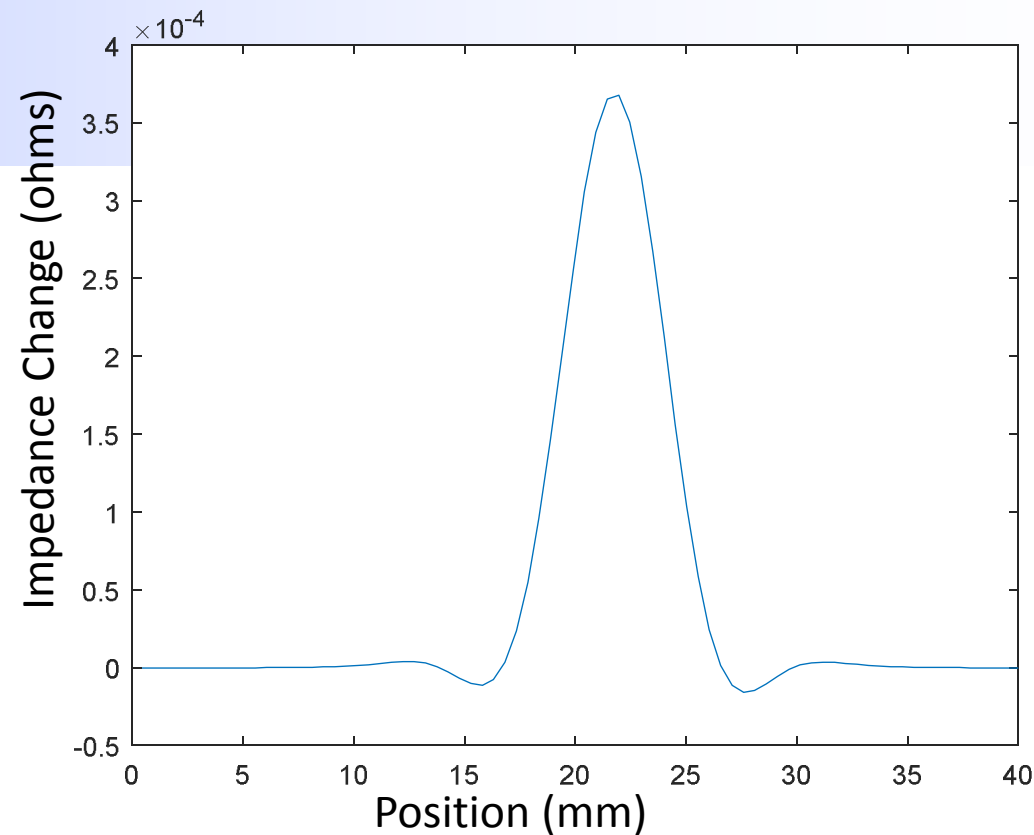
Angle = 1.3519 rad. Rotate to 2.9227 rad to remove

Reactance (ohms)

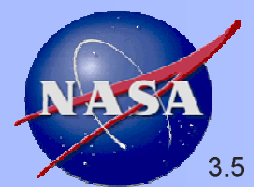
Nondestructive



Impedance Change (ohms)



Eddy Current C-scan, flaw response rotated vertical



Nondestructive

Channel 1

Angle = -1.3523 rad. Rotate to 0.21851 rad to remove

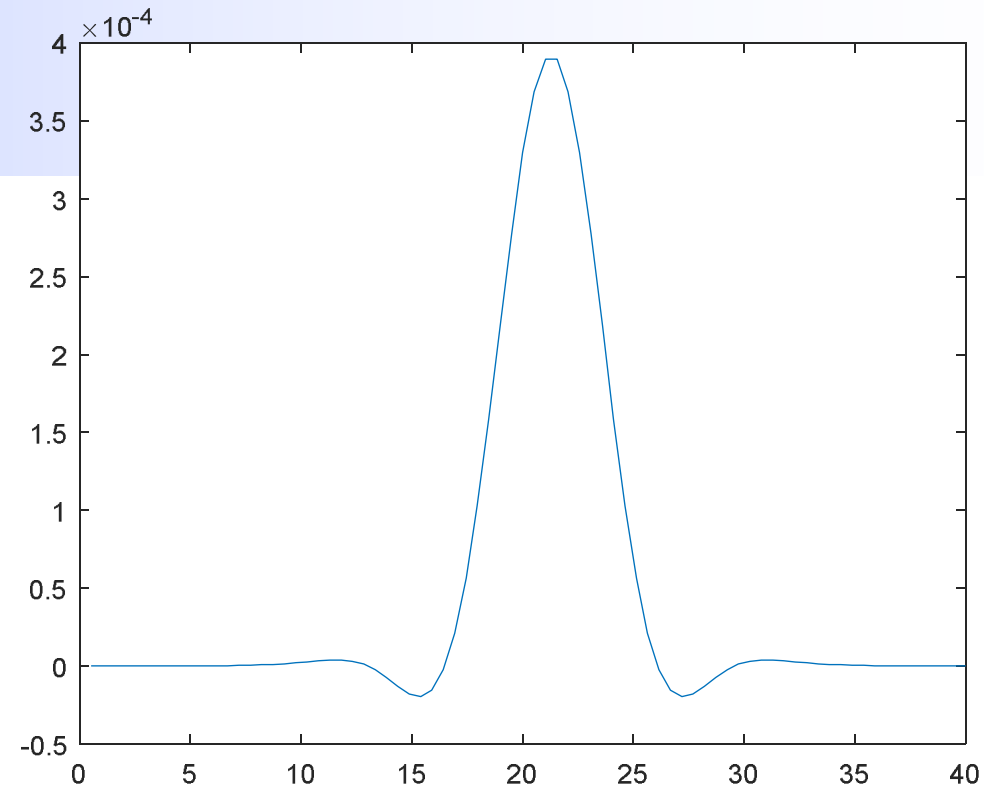
Reactance (ohms)



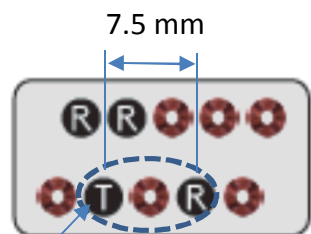
Resistance (ohms)

$\times 10^{-4}$

Impedance Change (ohms)

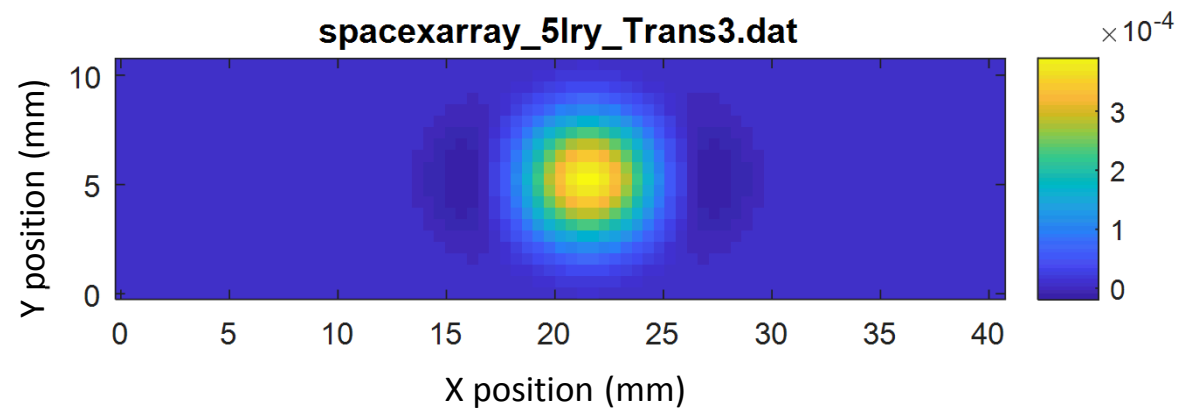
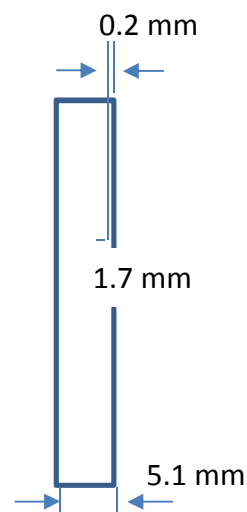
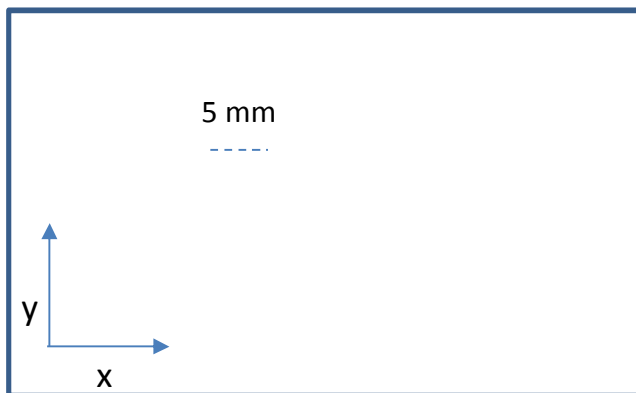


Position (mm)

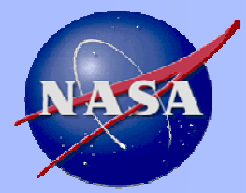


Long, single driver

3.5 mm dia

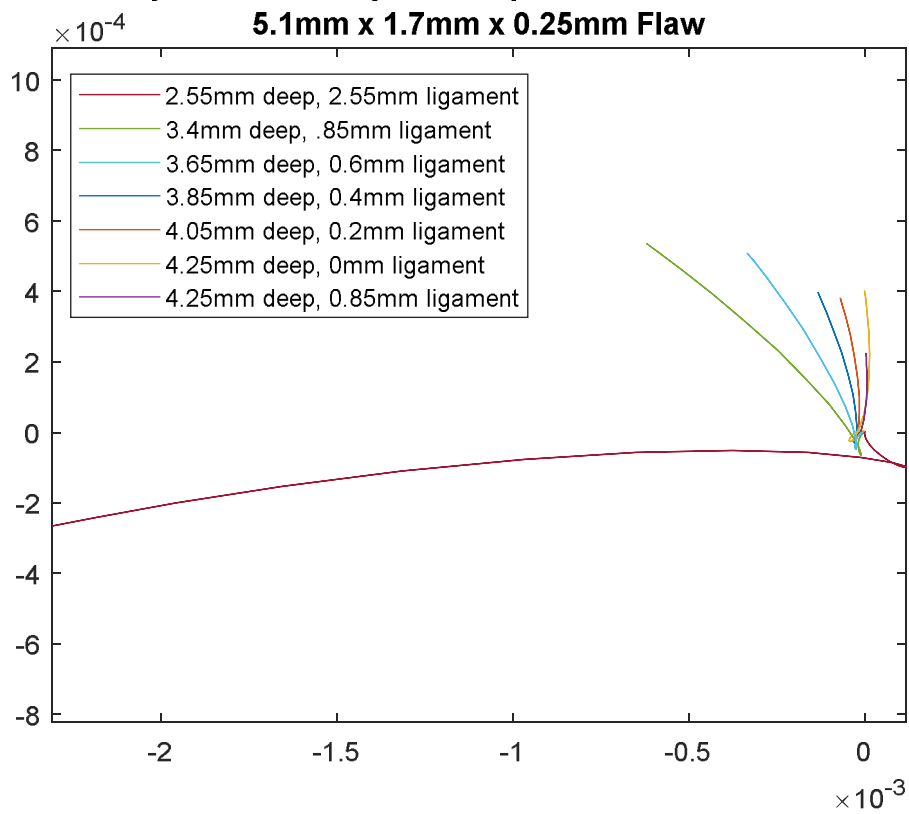


Eddy Current C-scan, flaw response rotated vertical

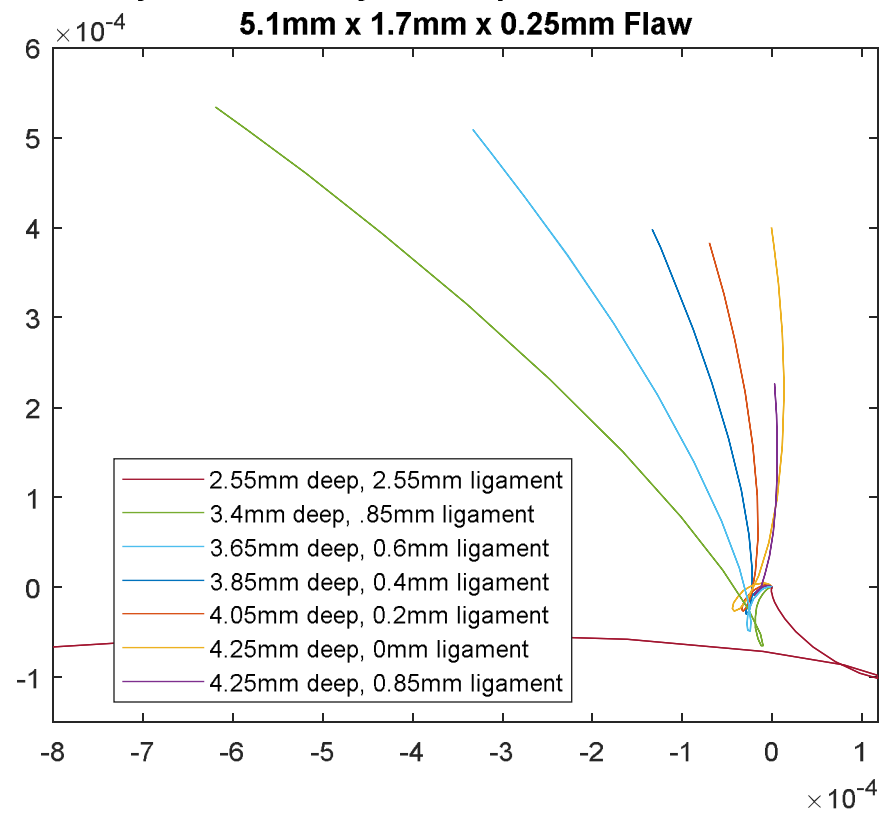


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Eddy Current Lissajous Response, Transverse Coils



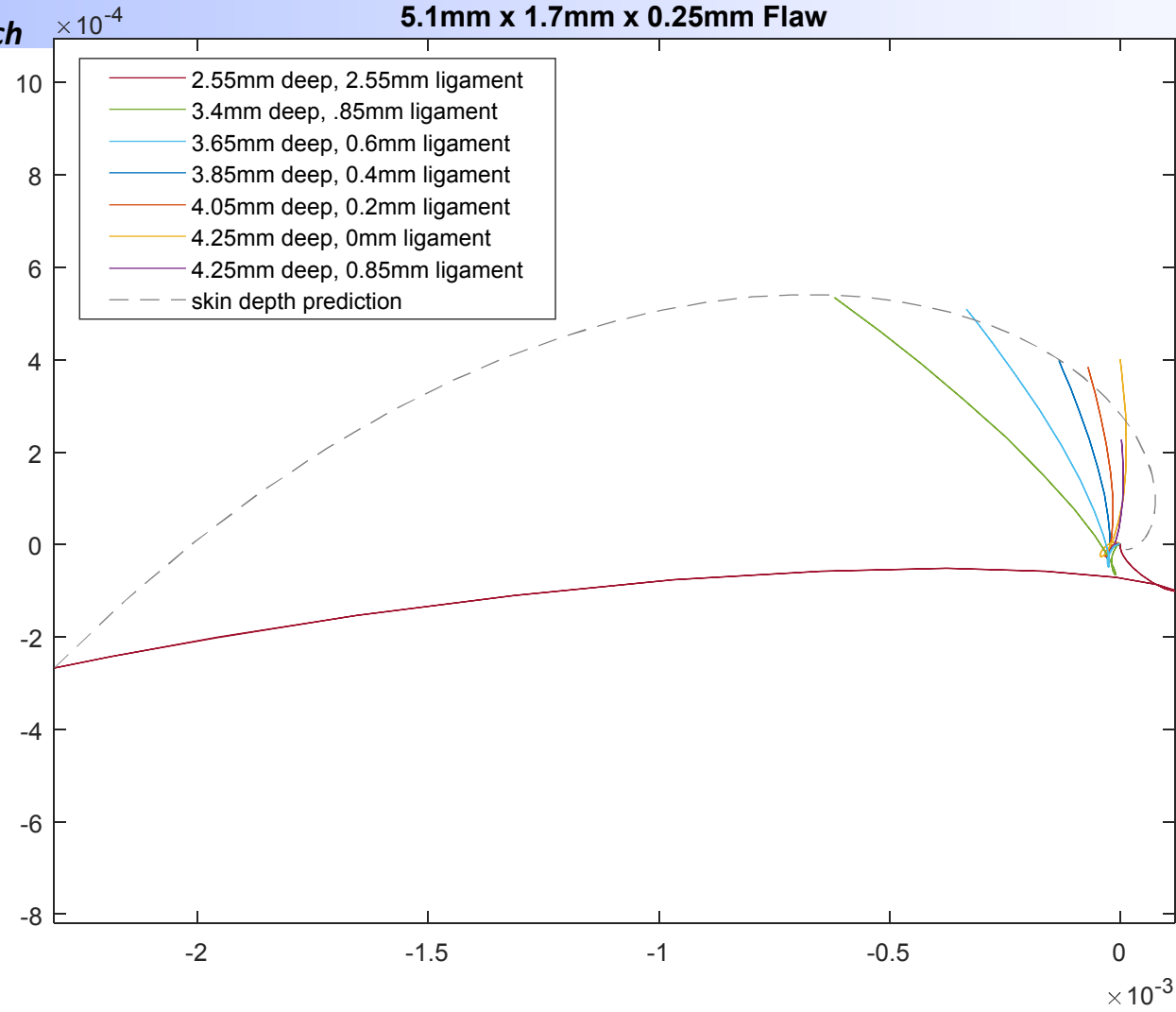
Eddy Current Lissajous Response, Transverse Coils

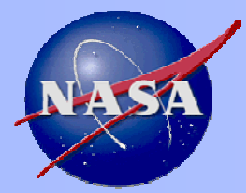




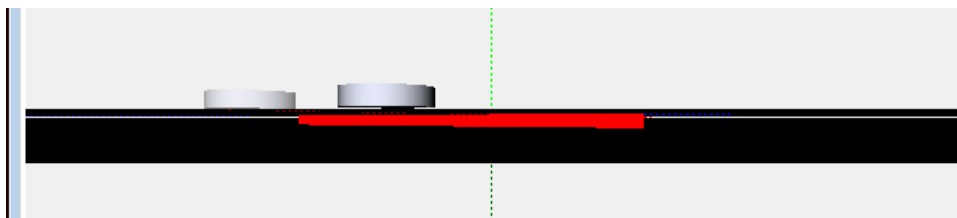
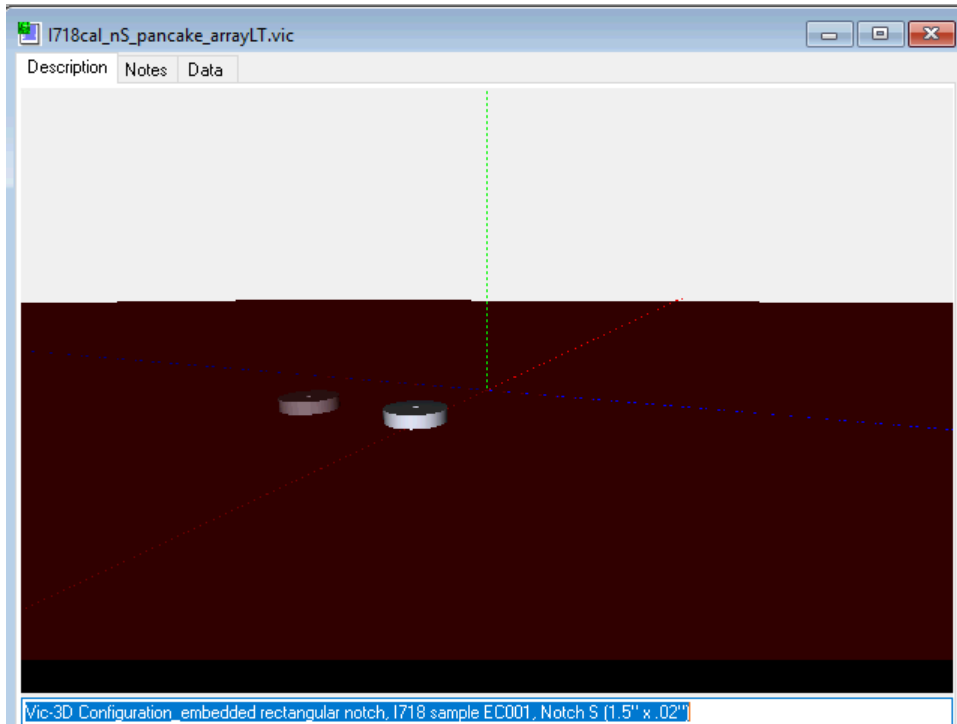
Nondestructive Evaluation Sciences Branch

Eddy Current Lissajous Response, Transverse Coils 5.1mm x 1.7mm x 0.25mm Flaw

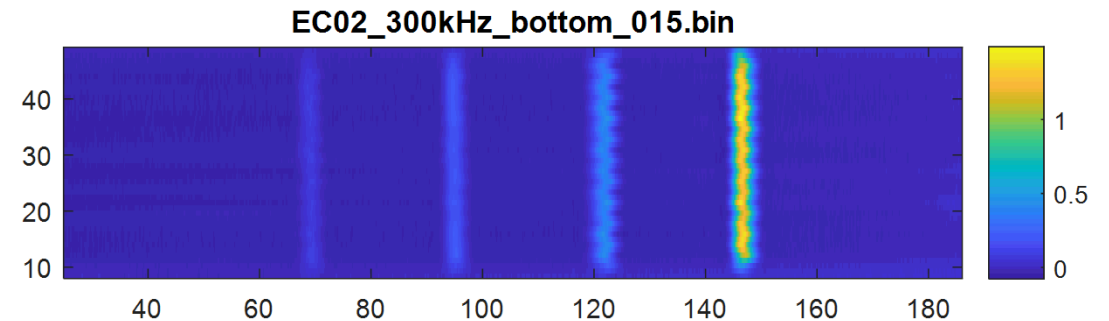




Modeling II, with Experimental Comparision



- Driver pickup coil pair modeled at 300 kHz, with coil diameter of 3.5mm and center to center coil separation of 7.5mm.
- Inconel 718 sample modeled with conductivity of 8×10^5 S/m and thickness of 0.050".
- Flaw modeled at 0.1mm wide notch and 25mm long, with varying depth and position in the sample.
- Experimental results obtained with Eddyfi Reddy system and ECA-IFG-056-050-048 array probe on 0.050" thick I718 calibration sample with 1.5" long x (.010, .015, .020, .030)" deep EDM notches.

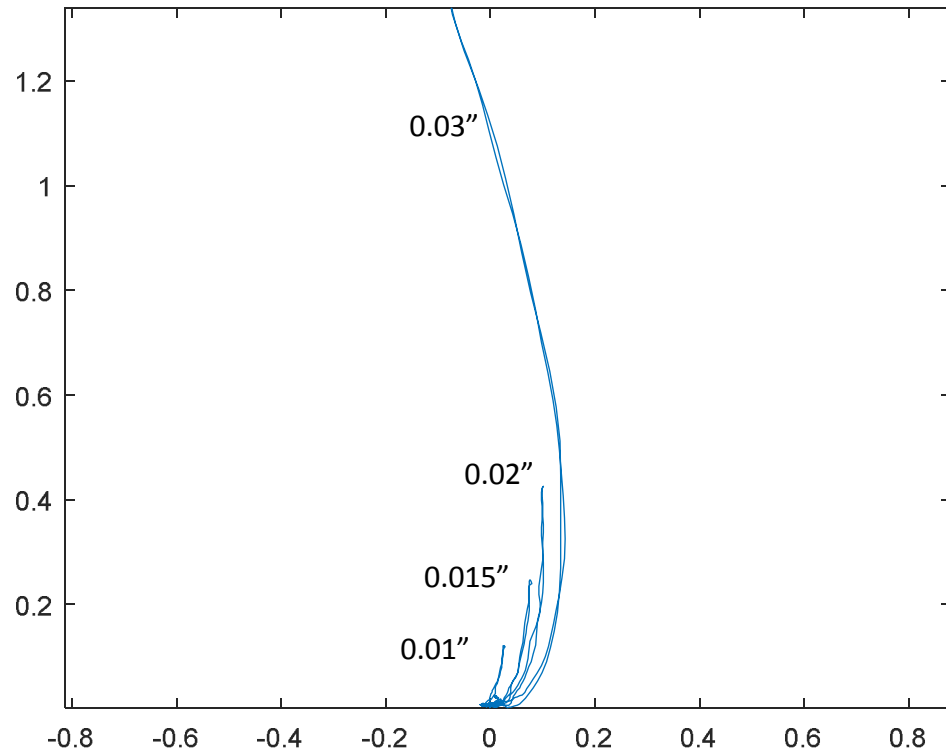




300KHz, ECA-IFG-056-050-048: Long Single Driver Transverse (LSDT) Configuration

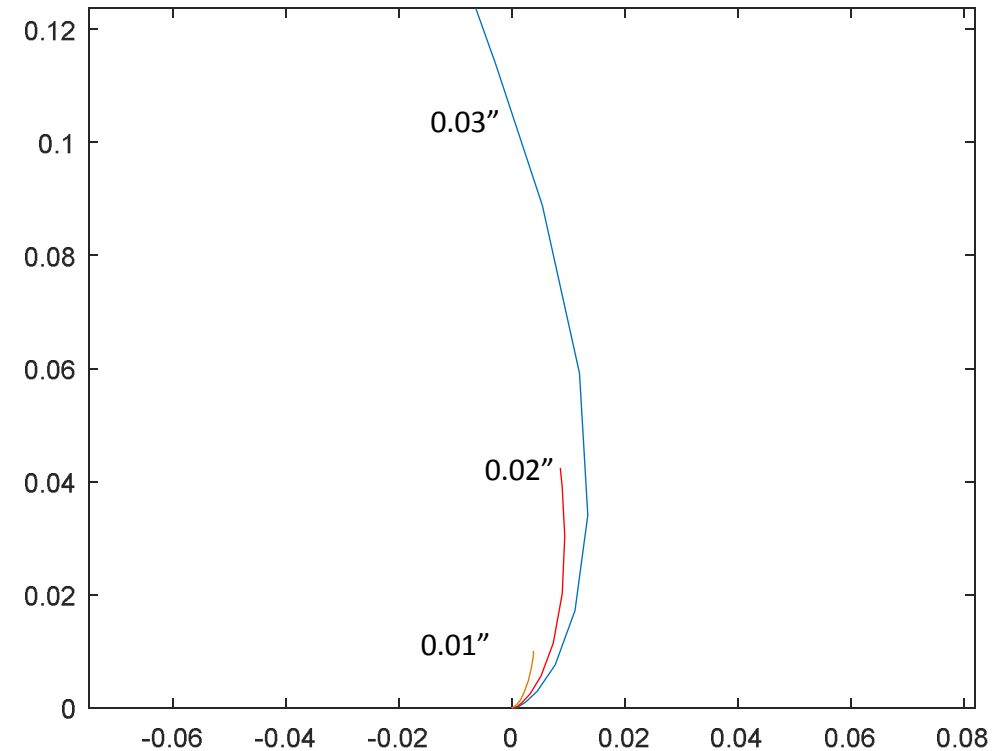
.01", .015", .02", and .03" deep flaws in back side of .050" thick I718 plate*

Lissajous response:
Phase Rotated Impedance Plane



Experimental Data

Lissajous response:
Phase Rotated Impedance Plane



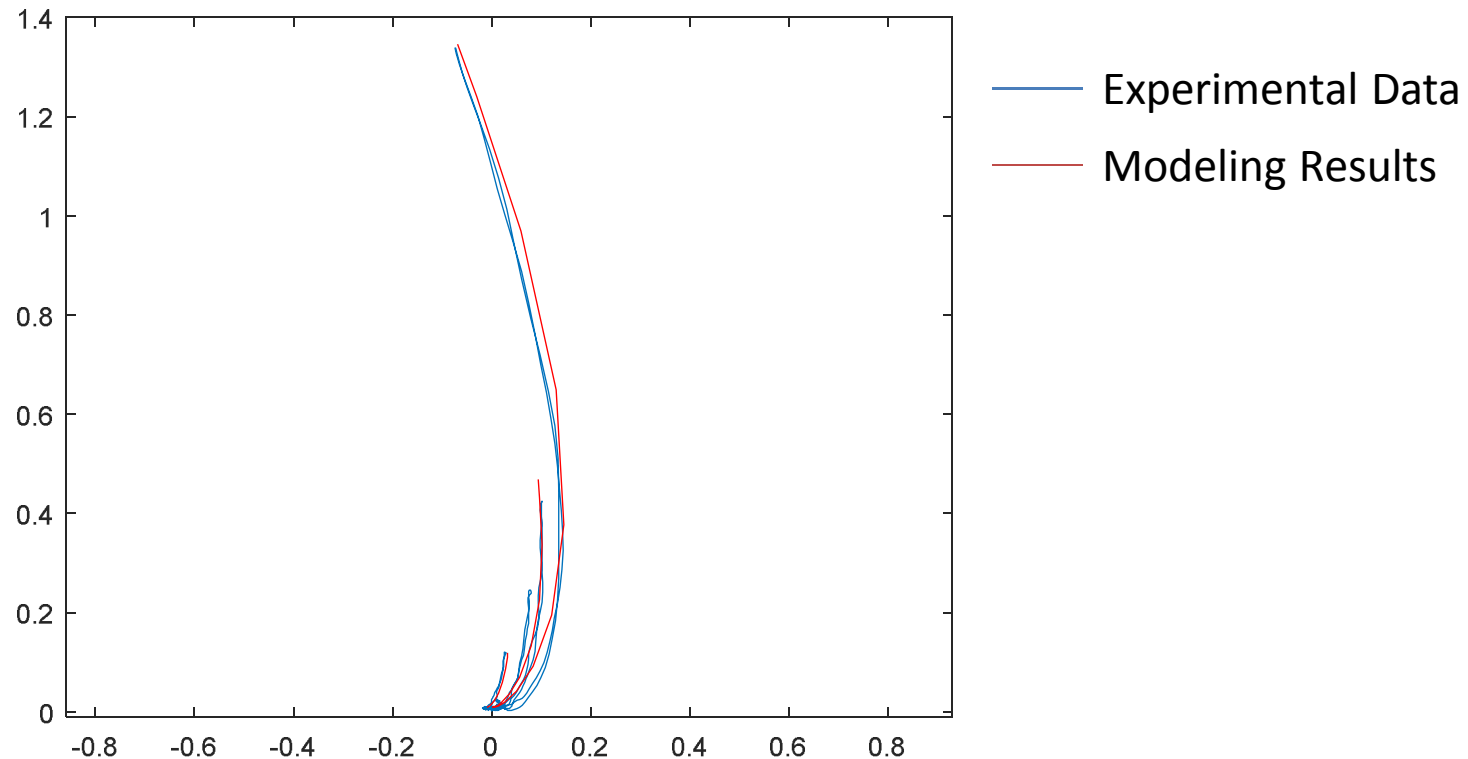
Modeling Results

*(.015" deep notch not modeled)



300KHz, ECA-IFG-056-050-048: Long Single Driver Transverse (LSDT) Configuration

.01", .015", .02", and .03" deep flaws in back side of .050" thick I718 plate*



Lissajous response: Phase Rotated Impedance Plane. Experimental results filtered and averaged. Modeling results scaled and rotated to match .030" deep notch with experimental results. *(.015" deep notch not modeled)

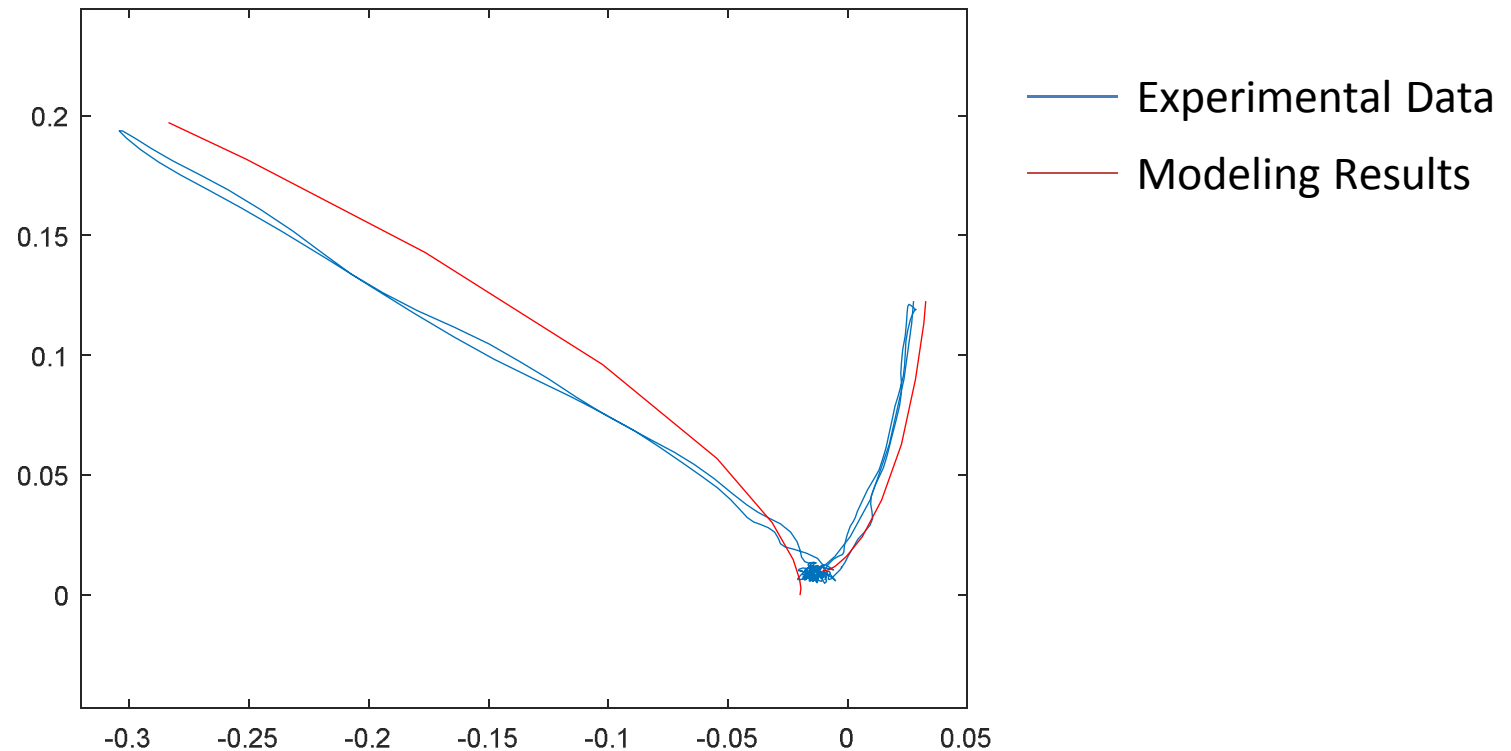


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300KHz, ECA-IFG-056-050-048: Long Single Driver Transverse (LSDT) Configuration

0.01" deep flaw at front and far surface in 0.050" thick I718 plate

Lissajous response:
Phase Rotated Impedance Plane



Front and Back Surface Breaking flaw results
Model data scaled and rotated as calibrated by back surface 0.030" deep notch.



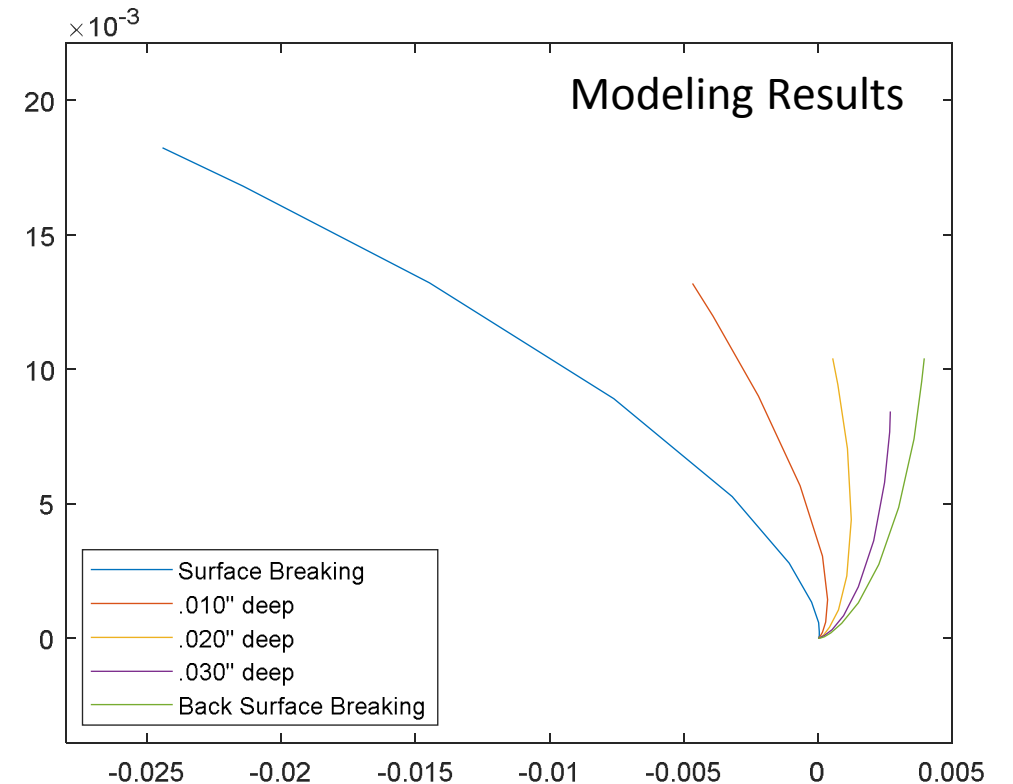
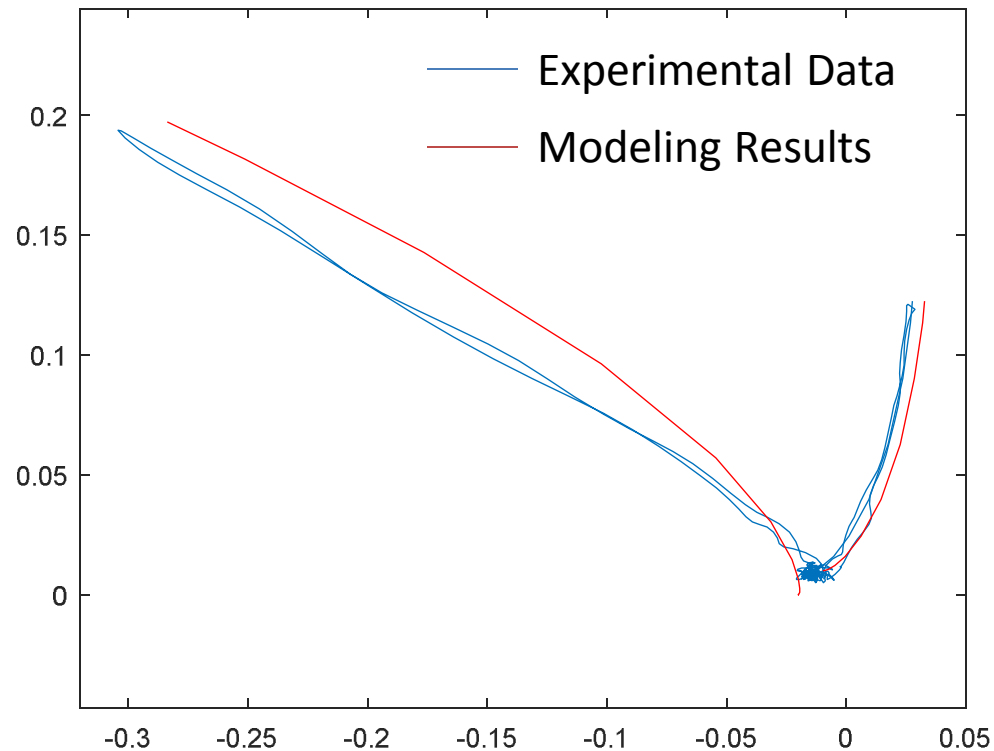
Nondestructive Evaluation Sciences Branch

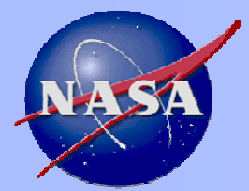
300KHz, ECA-IFG-056-050-048: Long Single Driver Transverse (LSDT) Configuration

0.01" deep flaw at varying depth in 0.050" thick I718 plate

Lissajous response:

Phase Rotated Impedance Plane

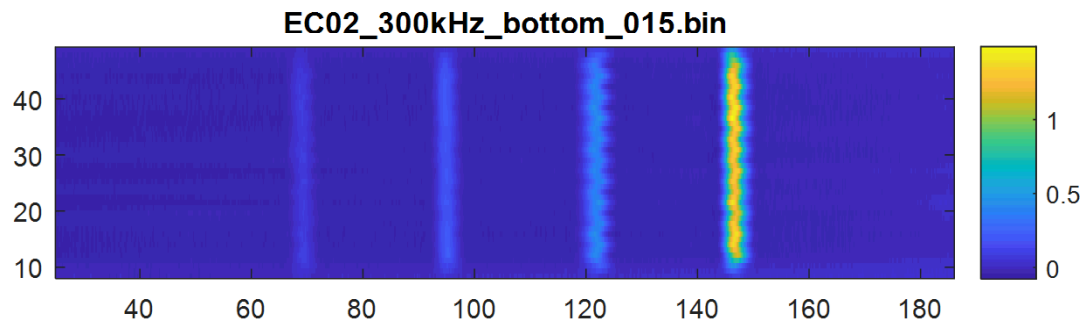




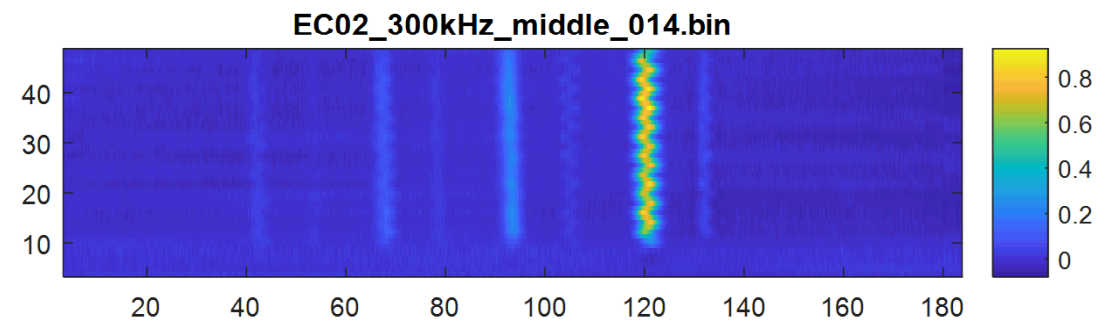
Nondestructive Evaluation Sciences Branch

300KHz, ECA-IFG-056-050-048: Long Single Driver Transverse (LSDT) Configuration

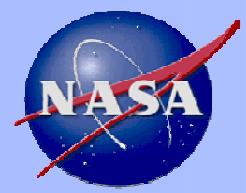
.01", .015", .02", and .03" deep flaws in 0.050" thick I718 plate



Single Layer, flaws on far surface



Double Layer, flaws at boundary

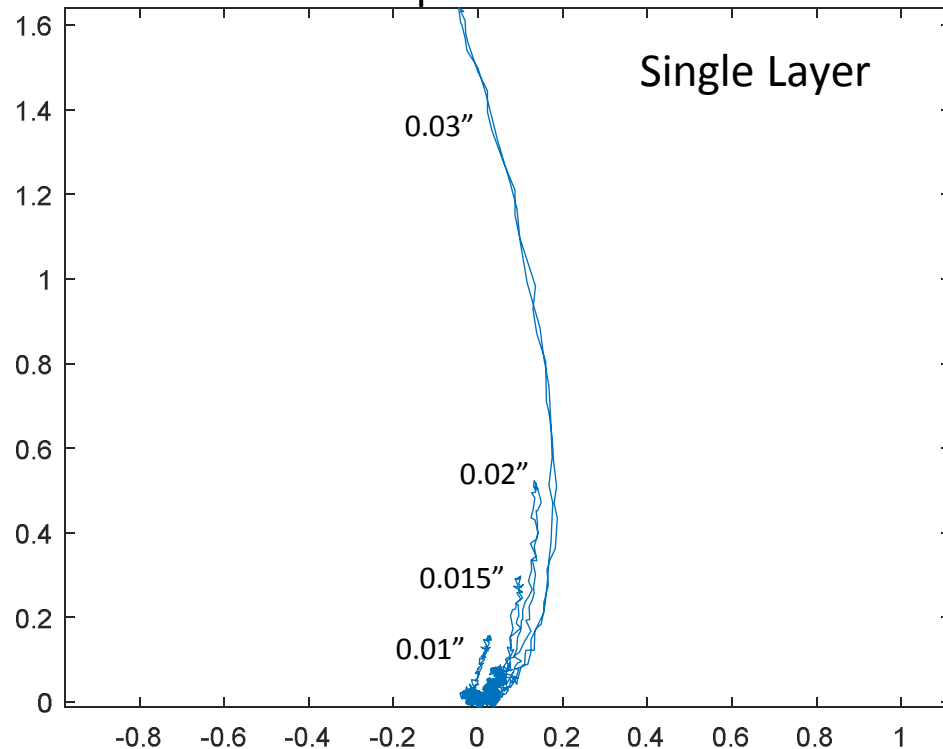


Nondestructive Evaluation Sciences Branch

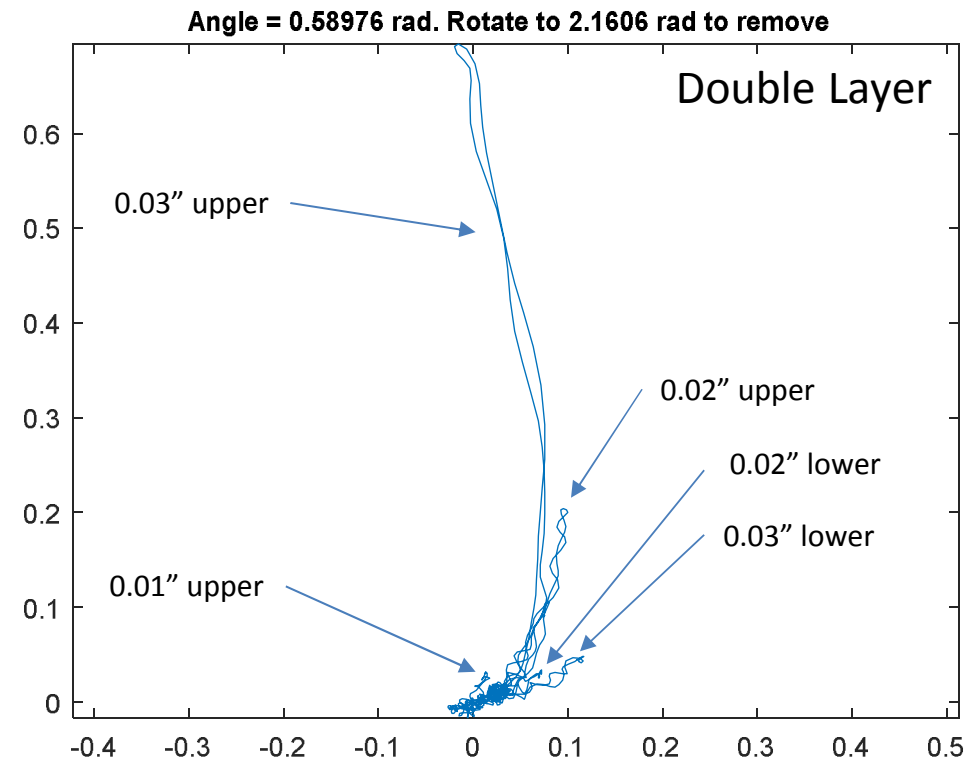
300KHz, ECA-IFG-056-050-048: Long Single Driver Transverse (LSDT) Configuration

.01", .015", .02", and .03" deep flaws in back side of .050" thick I718 plate

Lissajous response:
Phase Rotated Impedance Plane



Lissajous response:
Phase Rotated Impedance Plane
Channel 4



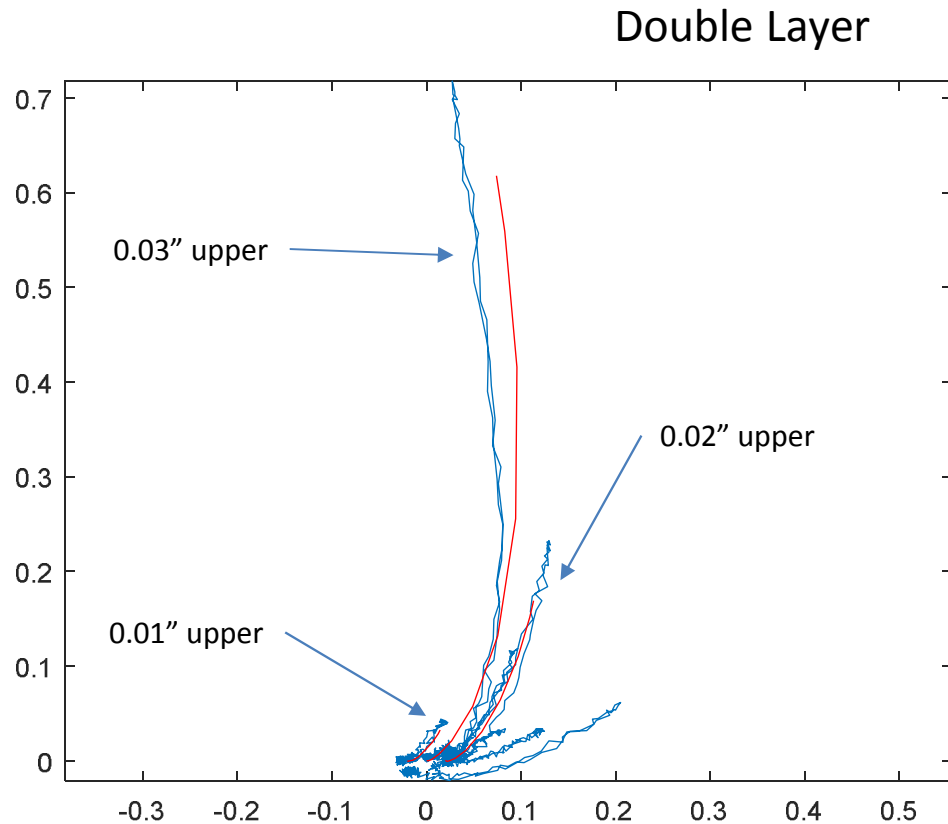


Nondestructive Evaluation Sciences Branch

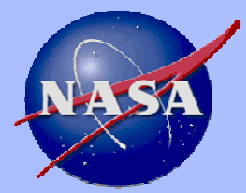
300KHz, ECA-IFG-056-050-048: Long Single Driver Transverse (LSDT) Configuration

.01", .015", .02", and .03" deep flaws at interface of two .050" thick I718 plate

Lissajous response:
Phase Rotated Impedance Plane

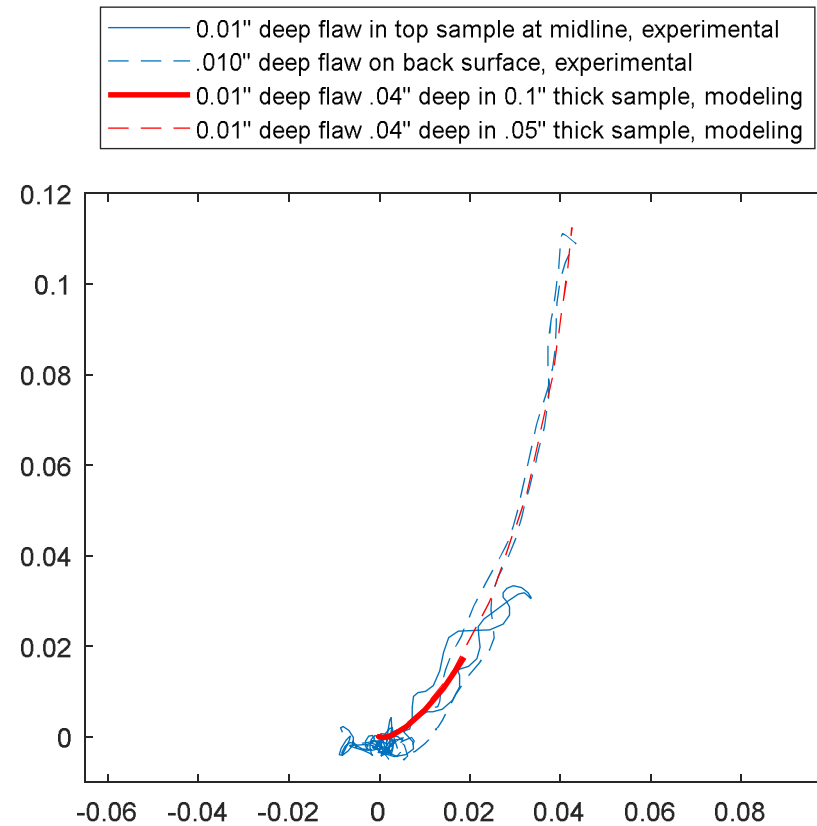


Model data rotated as per previous results.
Model amplitudes multiplied 2x above previous results



Comparison of Modeling and Experimental Results for Back Surface and Embedded Flaws

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Summary

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- Eddy Current techniques can be useful for the detection of fully embedded flaws when changing curvature, thickness, or layer boundaries limit the effectiveness of ultrasonic inspections.
- Care must be taken in calibration techniques and setting of flaw call thresholds.
 - Modeling results show that a standard utilizing a back-side flaw will overestimate the eddy current response as compared to an embedded flaw at the same depth.
 - The use of a backside flaw may still be conservative for the detection of a midline flaw in a structure as effects of induced current flow around the embedded flaw is balanced by signal loss due to skin depth penetration.
- Electromagnetic modeling can be used as an efficient technique to predict the response from embedded flaws.
 - Models can be experimentally tested for near and far surface flaws.
 - Model extrapolation to embedded flaws can be used to predict experimental response.
 - Techniques are in work for limited experimental confirmation on embedded defects.